Cleaner Firms, Dirtier Trade:

Carbon Offshoring and the Limits of Production-Based Accounting

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Abstract

Manufacturing in rich countries is getting cleaner—on paper. Recent firm-level studies find little evidence that offshoring to pollution havens drives this clean-up. Linking plant-level fuel use for Swedish manufacturers (2006–2014) to imports weighted by their embodied carbon, I instrument firms' carbon imports with a shift-share of foreign export-supply shocks. A 10 percent rise in carbon imports lowers production-based CO_2 intensity by 6.7 percent. This is not cleaner technique: emissions in levels barely move while value added rises. The pollution-haven margin, sourcing the same products from dirtier origins, is weak on average, as earlier studies find, but significant among firms that bought many distinct products before the shock, the only ones with room to reallocate. Yet the carbon does not disappear: firms' own transport emissions and the international freight carrying their imports both rise, and the national inventory attributes neither to Swedish manufacturing. The imported goods embody 64 percent more carbon than producing them in Sweden would. At most 2 percent of imported carbon enters through a country where the importing firm owns a plant, so leakage policy aimed at multinationals misses almost all of it, and the EU's carbon border adjustment, which excludes freight, misses the transport leg. Where emissions are counted can matter as much as whether they fall.

JEL: Q54, Q56, F18, F14, F23.

Keywords: carbon offshoring, carbon leakage, freight emissions, carbon footprint, shift-share, Swedish firms

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1 Introduction

A firm-level literature has asked whether offshoring moves dirty production to pollution havens, and has mostly answered that it does not. The verdict is recent, it is carefully drawn, and it is consistent across several countries. Yet in Swedish manufacturing the goods firms buy abroad carry 1.64 times the carbon that making them at home would have released. Both of those cannot be the whole story.

What reconciles them is what the accounts record. When a Swedish plant stops making a part and buys it instead, its own combustion falls, and the national inventory, the Emissions Trading System and, from 2026, the European carbon border adjustment all register that fall as abatement. None of the three records the carbon released making the part abroad, and none records the fuel burned carrying it to Sweden. A firm can therefore clean up on every ledger a regulator reads while the carbon it is responsible for has only changed address.

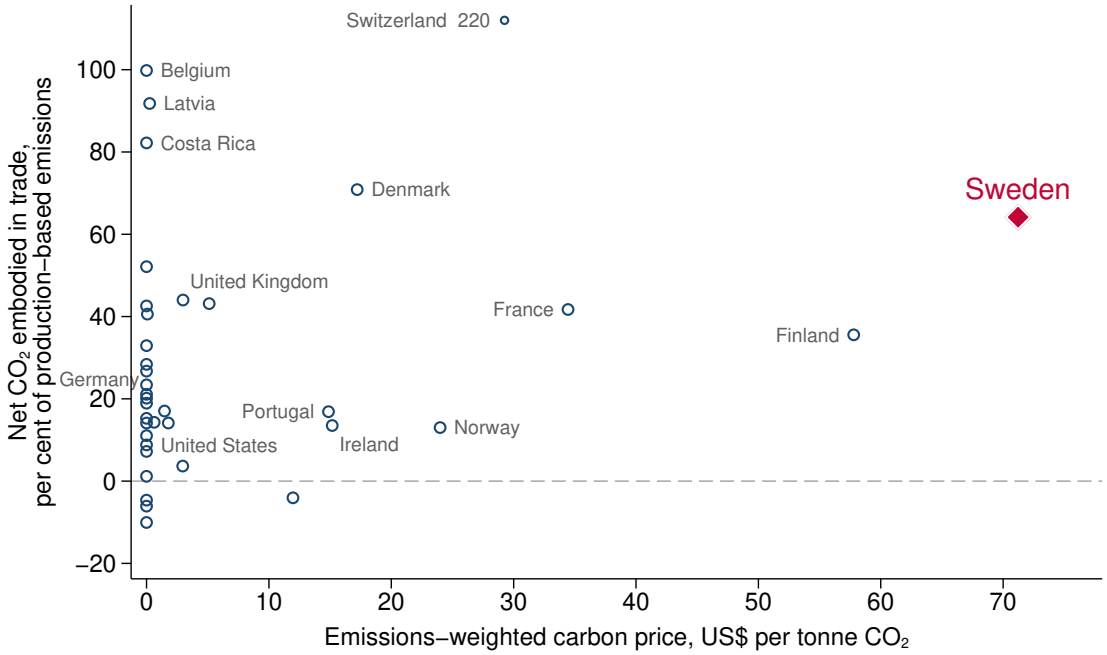
This paper measures that gap inside the firm. I link Swedish manufacturers' plant-level fuel use to their imports, weighted by the carbon embodied in them, and instrument those imports with movements in foreign export supply. A 10 percent rise in the carbon a firm imports lowers its emissions per unit of value added by 6.7 percent. The fall is not cleaner technique. It is less energy per unit of output, with part of it the denominator growing, not the numerator shrinking. On the haven question the literature has mostly answered no, and the pooled estimate here reproduces that answer; but it conceals the firms the mechanism runs through, because only a firm already buying many distinct products has anywhere to reallocate. And the carbon that leaves the ledger does not leave the world: the firm's own transport and the international freight carrying its imports both rise, and no inventory records either.

Sweden provides a useful setting for examining that divergence. Its manufacturers have been getting cleaner for two decades: between 2007 and 2014, air pollution emissions from Swedish manufacturing firms fell by 29 percent while the physical quantity of output rose by about 70 percent, and the United States (Levinson, 2010) and the rest of Europe (Brunel, 2017) show the same pattern. Sweden also prices carbon more aggressively than any other country, and is a large net importer of it: the emissions embodied in what it buys from abroad exceed those embodied in what it sells by nearly two thirds of everything its own territory emits.

There are two possibilities explaining manufacturing cleanups: *technique*, where firms use the same mix of goods but employ production processes that pollute less, and *composition*, where multi-product firms in developed countries can adjust their product mix towards low-emission-intensive and low-marginal-cost products. Previous research shows that an increase in climate

policy stringency in these jurisdictions can drive the use of energy-efficient equipment and better end-of-pipe controls, a phenomenon dubbed the "technique effect" (Levinson, 2023; Najjar and Cherniwchan, 2021; Brunel, 2017; Agostini et al., 1992; Andersson, 2019; Ponce and Khan, 2021; Marin and Vona, 2021). Far less is known about the composition effect, and almost nothing about one version of it: that the dirty step was not improved but bought from somewhere else. Which of the two accounts for Sweden's clean-up is the question this paper asks.

Figure 1: Sweden prices carbon hardest and still imports a great deal of it



Notes. OECD economies in 2020. The horizontal axis is the emissions-weighted carbon price, which multiplies the statutory rate by the share of a country's CO_2 it covers; the vertical axis is net CO_2 embodied in trade as a percentage of production-based emissions, so a positive value means a country consumes more carbon than it produces. Switzerland is plotted at the axis maximum and labelled with its true value. National accounts, constructed separately from the firm data used below. *Sources:* Global Carbon Project and the World Carbon Pricing Database, retrieved from Our World in Data; see `ourworldindata.org/grapher/share-co2-embedded-in-trade` and `ourworldindata.org/grapher/emissions-weighted-carbon-price`. Both series are documented at `ourworldindata.org/consumption-based-co2`. The extracts used here are committed with the replication files.

Figure 1 places Sweden against the other OECD economies on both margins at once. In 2020 its emissions-weighted carbon price was the highest in the world at 71 dollars a tonne, and its net carbon imports ran to 64 percent of its production-based emissions. The economies that import more carbon than Sweden (Switzerland, Belgium, Latvia, Costa Rica and Denmark) price it at a fraction of the Swedish rate, and those that price carbon comparably, Finland and France among them, import far less. No other economy combines the two. If stringent carbon pricing moves

dirty production abroad instead of removing it, Sweden is where the margin should be visible.

Thanks to free trade, a firm can import a polluting good it once made itself and concentrate what remains at home on less penalised products (Baumert et al., 2019; Jakob et al., 2022; Levinson, 2010; Nielsen et al., 2021). The idea is easy to state and has been hard to establish. Three things stand in the way, and they compound. Offshoring is chosen, so a firm that moves production abroad is not a random draw from the firms that do not. Emissions are recorded where combustion happens, so the carbon that leaves a country's manufacturing sector leaves its statistics at the same moment and cannot be followed. And the carbon embodied in a given import depends on the technology of the country that made it, which is not in the customs record. Each obstacle has pushed the literature toward aggregate data, where differences across countries in pollution-intensive production can be compared (Copeland and Taylor, 2004; Brunel, 2017; Levinson, 2010, 2023) but where the firm's own decision is no longer visible, and where the margin the hypothesis is about, one firm replacing its own dirty step with someone else's, cannot be separated from everything else moving at once.

The first obstacle is selection, and it is why a cross-section settles nothing. A firm adopting energy-efficient equipment may import more at the same time, which biases the association between offshoring and emissions toward zero; one in decline may do both less. I therefore instrument a firm's embodied carbon imports with a shift-share built on world export supply. For each EXIOBASE product and origin country I compute the growth in that origin's exports of that product to the rest of the world, excluding Sweden, and weight it by the share of the product in the firm's own imports in 2005, before the estimation window opens. The instrument moves the carbon embodied in what a firm already buys, for reasons originating in other countries' supply conditions, not in the firm's circumstances.

The exclusion restriction is defended from the shifts, not the shares (Goldsmith-Pinkham et al., 2020; Borusyak et al., 2022). Shares are plainly not randomly assigned, and I do not claim otherwise; what the design requires is that movements in world export supply are unrelated to shocks to Swedish firms' emission intensity once firm, sector-year and municipality-year effects are absorbed. One test disciplines that choice; no appeal to absorbing more variation does. The five-margin decomposition contains a placebo by construction: the emission intensity of *Swedish* production of the products the firm imports, which describes Swedish technology and cannot respond to one firm's sourcing. At firm and sector-year effects alone it moves, significantly. Adding municipality-year effects, it does not. I therefore prefer the specification with municipality-year effects, and report all three.

The second obstacle is that emissions are recorded where combustion happens. I decompose

the firm's carbon account across the boundary production-based accounting draws. The Swedish energy survey records physical quantities by fuel and by end use, so I separate what a firm burns in fixed equipment from what it burns in its own vehicles, and own combustion from purchased power. To these I add the international freight embodied in its imports, built from import weights, origin distances and mode-specific emission factors. I estimate all four on one sample with one instrument, so their movements are directly comparable.

That leaves the third obstacle, and it is the one I can do most about. Carbon offshoring is a statement about carbon, but what customs records is value, and the firm-level literature has had to proxy the one with the other. I measure it directly: each import line is weighted by the total emission intensity of its origin sector, direct plus all upstream, computed from EXIOBASE as $\mathbf{S}(\mathbf{I}-\mathbf{A})^{-1}$. Two firms importing a million kronor of inputs from Germany and from China are then no longer treated as having offshored the same thing, because they have not. The regressor is the quantity the hypothesis is about.

The estimated elasticity is -0.670, and just over two thirds of it is the denominator. On the same specification value added rises by 0.463 and sales by 0.406, both precisely estimated. The rest is not a change in technique. On the strongest first stage in the paper no fuel share responds and emissions per gigajoule do not move, and these firms already draw 77 percent of their energy as electricity, so there was little dirty fuel left to switch away from. What falls is energy per unit of output.

Under the same instrument and the same specification the elasticity is -0.670 when the regressor is embodied carbon and -0.785 when it is trade value. Both are precisely estimated. The weighting does not manufacture the result; it changes what the result is about. Import value cannot say where the carbon went, or how much of the aggregate decline it accounts for, because carbon is the object being counted.

Estimating every component on the same instrument, the only ones that respond are the ones no inventory records. The firm's own transport fuel rises, and so does the international freight carrying its imports. Production-based accounting is constructed to exclude both: own transport because inventories attribute it to the transport sector, freight because it burns on a carrier's books in no manufacturing country's account. Freight is a median 2.4 percent of the carbon embodied in a firm's imports, so its magnitude is not comparable with the domestic reduction. Its sign runs against the firm.

Whether that carbon moved to dirtier producers is the pollution-haven question, and the firm-level literature has mostly answered no. I find that it moved, for the firms in a position to move it. Embodied carbon is the product of three things: how much a firm buys, what it buys, and

how dirty the places it buys from are. I estimate the response of each. Volume and product mix carry most of it. Sourcing the same goods from dirtier origins is weak on average, as earlier work reports, and significant among the firms that bought many distinct products before the shock arrived. The framework predicts that pattern: with a fixed cost per origin, only a firm already sourcing widely has room to reallocate. The pooled null averages over firms with no such margin, so it is uninformative about the firms that have one. I ask which margin of embodied carbon the shock moves because the instrument will not be split by origin group, which Appendix F sets out.

The carbon arrives through arm's-length trade, not through ownership. That distinction matters because the pollution-haven hypothesis has been tested almost entirely on foreign direct investment, the channel aggregate data record. In firm records the intra-firm channel is too small to carry the result: at most 2.0 percent of imported manufacturing carbon enters through a country where the importing firm owns a plant, and that figure is an upper bound, not an estimate. Nor does ownership make sourcing dirtier. Within a firm, across the origins it buys from, owning a plant raises how much it sources and leaves carbon per krona unchanged to within two percent. And opening a first affiliate abroad does not move the parent's emission intensity over the horizons a panel ending in 2014 can follow. Relocation is slow, so that bounds a fast ownership response and leaves a slow one open. Over the decade I observe, the margin that moved was trade. A leakage policy keyed to who owns the foreign plant would reach a small corner of the flow.

Related literature. A firm-level literature establishes that imports and offshoring lower manufacturing emission intensity, including in Sweden (Akerman et al., 2024; Dussaux et al., 2023; Li and Zhou, 2017; Choi et al., 2023). On the haven question its recent verdict is negative: Akerman et al. (2024) find no pollution-haven effect in Sweden, Dussaux et al. (2023) none in France, Leisner et al. (2026) none in Denmark, and Forslid et al. (2018) an anti-haven pattern among Swedish exporters. I reconcile that verdict instead of contesting it. The haven margin is detectable, but only among firms with the product breadth to reallocate, and not through the ownership channel on which the hypothesis has usually been tested. Akerman et al. (2024) use the same Swedish registers and attribute the import–emission link largely to cleaner technology; I locate the masking elsewhere, in emissions outside the production boundary. Shapiro and Walker (2018) attribute most of the United States manufacturing clean-up to the implicit pollution tax and not to trade, and this paper is the firm-level counterpart on the trade side.

Two choices separate the design from its closest antecedents, one in the regressor and one in the outcome. Separating genuine offshoring from broader import competition has been the

central difficulty in this literature (Leisner et al., 2026; Lehr, 2025), and I make that distinction operational by weighting each import line on its origin's carbon content instead of its trade value, scaling by value added as in Cherniwchan et al. (2017) because emissions per krona of revenue fall almost mechanically when a firm moves tasks off site. On the outcome side, Cherniwchan (2017) finds that NAFTA tariff cuts account for roughly two thirds of the fall in United States manufacturing emissions of particulates and sulphur dioxide between 1994 and 1998, and attributes that fall to declining emission intensity. My estimates run the other way on mechanism: emissions in levels do not move, value added rises, and the intensity decline is accordingly a denominator effect. Emissions from moving goods have in turn received much less attention than emissions from making them (Copeland et al., 2022). I build firm-level freight carbon following Cristea et al. (2013), with modal factors from the GLEC Framework and the IMO Fourth Greenhouse Gas Study, and find it to be the one component of the account that responds positively to the shock.

2 Conceptual Framework

A country's emission inventory is a boundary drawn around its territory, and a firm's carbon does not respect it. When a Swedish manufacturer buys a dirty input abroad instead of making it, carbon leaves the Swedish ledger in three distinct ways, only one of which is a reduction. This section writes that down, and then derives, from the firm's sourcing problem, the margins along which it happens, so that what the paper later estimates is a set of comparative statics of a single decision.

Let M_{ict} be the quantity of inputs firm i imports from country c and $M_{it} = \sum_c M_{ict}$ its total imports, e_{ct} the carbon intensity of production there, d_c the shipping distance, and $\varphi_{m(c)}$ the emission factor of the transport mode used on that route. The firm's total carbon footprint is

$$\underbrace{E_{it}^{\text{TOT}}}_{\text{footprint}} = \underbrace{E_{it}^{\text{DOM}}}_{\text{domestic production}} + \underbrace{\sum_c M_{ict}\, d_c\, \varphi_{m(c)}}_{\equiv E_{it}^{\text{FR}}\ \text{(international freight)}} + \underbrace{\sum_c M_{ict}\, e_{ct}}_{\equiv E_{it}^{\text{FGN}}\ \text{(embodied abroad)}} . \tag{1}$$

The last term is the carbon-import measure used throughout the paper. National inventories and the firm-level literature measure only the first.[1]

The gap between the two is not a rounding error, and its sign is known in advance. Offshoring

[1] Inventories assign emissions to the country where combustion occurs. Domestic transport fuel burned by the firm is part of E^{DOM}; international freight is attributed to the carrier's bunkers and to the exporter's territory, so it appears in no importing firm's production-based footprint.

a unit of production lowers E^{DOM} by the domestic clean-up it achieves, raises E^{FGN} one for one, because the embodied measure *is* the thing being offshored, and raises E^{FR} by whatever it costs to ship. Production-based accounting records the first movement and none of the second or third. The cleaner a firm looks in the inventory, therefore, the more carbon it may have relocated and shipped, and whether its total footprint falls at all depends on a race between the domestic saving and what replaces it. That race could be summarised in a single leakage rate. The decomposition set out below is the more informative object, because it says which part of the race is moving.

2.1 The sourcing problem

The components of (1) are not free parameters. They come from a sourcing problem of the kind in Antràs et al. (2017), in which a firm chooses which countries to buy from subject to a fixed cost per origin, and in which sourcing enters through the cost function and not additively across markets, so that the decisions interact. That interaction is what later generates a prediction about *which* firms adjust on which margin, and no accounting identity can produce such a prediction.

A firm i produces using inputs indexed by product $p \in \mathcal{P}_i$, bought at home or from a country in its *sourcing set* Ω_i, where establishing a relationship with country c costs $f_c > 0$ per period. Country c offers product p at effective cost $w_{ct}\tau_c / a_{pc}$, with Fréchet productivity draws of shape θ. The firm's sourcing capability is then $\Theta_{it} = \sum_{c \in \Omega_i} T_c (w_{ct}\tau_c)^{-\theta}$, the unit cost of its input bundle is proportional to $\Theta_{it}^{-1/\theta}$, and the share of product p's spending going to country c is

$$s_{ipct} \;=\; \frac{T_c \left(w_{ct}\tau_c\right)^{-\theta}}{\Theta_{it}}, \qquad c \in \Omega_i. \tag{2}$$

This is independent of p: origin shares are common across products within the sourcing set, while product shares ω_{ipt} are fixed by the firm's technology. That separation is what lets the origin margin and the product margin be told apart empirically, and it is an assumption; I return to it below.

Writing the embodied carbon of imports as $E^{\text{FGN}}_{it} = M_{it} \sum_p \omega_{ipt} \sum_{c \in \Omega_i} s_{ipct} e_{pc}$, taking logs, and adding and subtracting what the same basket would have emitted at Swedish intensities e^{SE}_{pt} (in the base year, e^{SE}_{p0}) gives the decomposition the paper estimates:

$$\ln E^{\text{TOT}}_{it} = \underbrace{\ln M_{it}}_{\text{VOLUME}} + \underbrace{\ln \textstyle\sum_p \omega_{ipt} e^{\text{SE}}_{p0}}_{\text{PRODUCT MIX}} + \underbrace{\ln \frac{\sum_p \omega_{ipt} e^{\text{SE}}_{pt}}{\sum_p \omega_{ipt} e^{\text{SE}}_{p0}}}_{\text{DOMESTIC TECHNIQUE}} + \underbrace{\ln \frac{\sum_p \omega_{ipt} \sum_c s_{ipct} e_{pc}}{\sum_p \omega_{ipt} e^{\text{SE}}_{pt}}}_{\text{ORIGIN DIRTINESS}} + \underbrace{\ln\!\left(1 + \tfrac{E^{\text{FR}}_{it}}{E^{\text{FGN}}_{it}}\right)}_{\text{FREIGHT}}. \tag{3}$$

Being an identity, it tells us nothing by summing. Its content is that the five terms respond

differently, and predictably, to one and the same disturbance. Which disturbance matters, so I take it next.

2.2 Comparative statics

Suppose foreign supply costs fall in a subset of origins, $d \ln w_{ct} < 0$ for $c \in \mathcal{S}$, for reasons having nothing to do with any Swedish firm. This is the disturbance, and Section 5 builds an instrument that isolates it. What matters here is only its character: it is an *origin* shock, changing the relative attractiveness of countries and not of products. That is what signs the margins, and they are not all signed the same way.

Volume is unambiguous. Cheaper inputs raise Θ_{it}, lower the unit cost of the bundle and raise import spending. Product mix is not: the shock does not enter ω_{ipt} directly, and product shares move only through the second-order channel by which cheaper inputs alter what the firm chooses to produce. Because ω_{ipt} is a technology object, it should also be indifferent to how broad a firm's sourcing set happens to be. Freight is proportional to import volume at a given distance composition, so unless the shock systematically favours distant origins over nearby ones, the carbon of moving goods and the carbon embodied in them scale together and the *wedge* between them does not move, even as freight in levels rises. Domestic technique, finally, is a property of Swedish production that nothing in the firm's sourcing problem touches, which makes it a placebo: if it moves with the shock, the estimation is picking up industry- or time-varying conditions that it ought to have absorbed.

Origin dirtiness is where the fixed cost bites, and where the firms differ. Consider a carbon-intensive origin $c' \notin \Omega_i$ that becomes cheaper. The firm adds it only if the gain exceeds $f_{c'}$. Because inputs enter through the cost function, that gain is proportional to total input spending, which scales with the breadth of the firm's product set: an origin serves every product the firm buys, so a broader firm amortises $f_{c'}$ over more uses. This is the complementarity across source markets that Antràs et al. (2017) emphasise, and it splits firms in two. A firm with a large Ω_i already holds the dirty origin and responds on the intensive margin, reallocating shares within Ω_i, which moves origin dirtiness immediately. A firm with a small Ω_i sits at a corner: responding at all means paying $f_{c'}$, and for shocks below that threshold it does not respond.

2.3 Predictions

The model yields three predictions, which Section 7.4 takes to the data in turn.

Prediction 1 *An origin-driven shock moves the product-mix margin only to second order. Its elasticity*

is small relative to the volume margin, and it does not differ systematically across firms with different sourcing sets, because ω_{ipt} is a technology object, not a sourcing object.

Prediction 2 *The freight* wedge $\ln(1 + E^{\text{FR}}/E^{\text{FGN}})$ *is invariant to the shock, even though freight emissions in* levels *rise with it. A zero on this margin is therefore not evidence that offshoring is transport-free; it is evidence that shipping scales with the carbon it carries.*

Prediction 3 *The origin-dirtiness response is increasing in the breadth of the firm's baseline sourcing capability. Firms with narrow product scope, and hence small sourcing sets, show no origin response; firms with broad scope show a positive one. The product-mix response, by contrast, does not vary with scope.*

The third is the one that discriminates, because it is a *differential* prediction: it says the two composition margins should part company across the same partition of firms. A story in which larger or more import-dependent firms simply respond more to everything would move both, and would move volume with them.

2.4 Assumptions

The origin shares in (2) are common across products within a sourcing set. This is what makes the origin and product margins separately identified, and it is an assumption: if a firm sourced steel from China and chemicals from Germany for reasons specific to each product, the two margins would not decompose cleanly. The data speak to this only indirectly, through the finding that basket composition at EXIOBASE resolution barely moves.

The model is partial equilibrium. Origin cost shifters w_{ct} are taken as given, which is the assumption the shift-share design makes in any case, and prices of Swedish output do not adjust. It is therefore silent on welfare. Finally, f_c is a fixed cost of sourcing *capability*, not of ownership: the model says nothing about whether a firm serves an origin through an affiliate or at arm's length, which is consistent with the finding in Section 6 that the intra-firm margin is too small in Swedish manufacturing to be measurable.

3 Data

Measuring carbon offshoring at the firm level requires three objects that are rarely observed together: a firm's own emissions, the full product-by-origin composition of its imports, and the

emission intensity of production in the countries it buys from. I assemble them from Swedish administrative microdata held by Statistics Sweden (SCB), matched to a global multi-region input–output table. This section describes each in turn, then the sample.

3.1 Data sources

SCB collects annual energy use for every manufacturing plant with at least ten employees. Firms report physical quantities by fuel (litres of diesel, MWh of electricity, hundreds of cubic metres of natural gas), not emissions. I convert quantities to CO_2 using fuel-specific combustion factors,

$$E_{it} = \sum_f \alpha_f \, x_{ift}, \tag{4}$$

where x_{ift} is firm i's consumption of fuel f in year t and α_f the CO_2 content of that fuel. The same construction is used by Forslid et al. (2018) and Barrows and Ollivier (2021).[2] Because α_f is a constant, (4) measures *stationary combustion*: the CO_2 released when the firm burns fuel on its own premises. It is not a full scope-1 inventory, since it excludes process emissions from chemical reactions such as calcination in cement or reduction in steel. For Sweden over 2005–2014 this is a modest omission for most of manufacturing but not for all of it, and I note below where it binds. The same energy data give a second, coarser measure of environmental performance, energy supply in gigajoules using calorific values, which is available for every fuel including those whose units I cannot reconcile across the two files. Energy and CO_2 answer different questions. Energy use is local in its incidence and varies sharply with how dirty the fuel mix is, whereas CO_2 is a global pollutant that crosses borders and persists (Parry et al., 2014); comparing them shows whether offshoring moves the local or the global margin. Energy is the primitive and CO_2 the derived object: $\ln(CO_2/VA) = \ln(E/VA) + \ln(CO_2/E)$, an identity I exploit in Section 6.4 to separate how much energy a firm uses from how dirty that energy is.

Holding α_f fixed assumes the carbon content of a fuel does not vary with abatement technology. For Sweden in this period that is reasonable: end-of-pipe carbon capture was not deployed at commercial scale. Second, emissions in levels confound environmental performance with firm size. I therefore work throughout with emission *intensity*, emissions per unit of value added, following Holladay (2016), Rodrigue et al. (2024) and Barrows and Ollivier (2021). No denominator is innocent. Value added and sales respond to prices, which carry no environmental content, while physical output requires units to be comparable across products and time and so covers

[2] SCB uses different fuel names in the energy-use file and the emission-factor file. I map them by hand and take the modal coefficient within each energy identifier, which absorbs variation in spelling. Factors are combustion-only and exclude life-cycle emissions.

fewer firms.

I match the energy data to three further SCB registers. The Structural Business Statistics (FEK) provide employment, industry (NACE Rev. 2), sales, value added, and balance-sheet items. The International Trade in Goods (ITG) register records the value and quantity of every import and export line by CN8 product and partner country. The Production of Commodities (IVP) register gives production quantities and values at the 8-digit product level. I keep NACE 10–33 (manufacturing) and the years 2005–2014, using 2005 as the pre-sample year from which instrument shares are taken.

Offshoring is not the same as importing. A firm that buys finished goods for resale is not relocating production, and counting all imports conflates production inputs with final goods and with goods never made domestically (El-Sahli et al., 2022). I therefore count only intermediate inputs, classified using the fifth revision of the Broad Economic Categories (BEC codes 121, 220, 420, 521, 530). Because BEC revisions reclassify products over time, I harmonise codes across revisions using the concordance algorithm of Pierce and Schott (2012) as extended by Van Beveren et al. (2012).

I keep NACE 10–33 not only because imports originate mostly there but because dirty industries are systematically more trade-exposed, facing lower tariffs and non-tariff barriers (Shapiro, 2021).

3.2 The carbon content of imports

The object of interest is not the value of these imports but the carbon embodied in them. Between 22 and 35 percent of global emissions are embodied somewhere in the value chain of traded goods (Copeland et al., 2022), and that share has been rising (Eaton et al., 2016), so the intensive margin of the toxic content of each import line is what matters, not its value (Dussaux et al., 2023). I construct

$$CarbImp_{it} = \sum_{p} \sum_{c} M_{ipct}\, EI_{pct}, \tag{5}$$

where M_{ipct} is firm i's import of product p from country c and EI_{pct} is the emission intensity of producing p in c.

The intensities EI_{pct} come from EXIOBASE, a multi-region input–output table covering 44 countries and 5 rest-of-world regions at 200 products. I use *embodied* intensities, $\mathbf{S}(\mathbf{I}-\mathbf{A})^{-1}$, which count emissions generated anywhere in the upstream chain of a product and not only at the final stage. Intensities are industry-by-country, not firm-by-product, because producer-level intensities for source countries do not exist; this is the standard proxy in the literature (Levin-

son, 2010; Li and Zhou, 2017), and it is a source of measurement error I return to in Section 5. Import lines are matched to EXIOBASE products through an HS-to-EXIOBASE crosswalk and to EXIOBASE regions through a country bridge; intensities are expressed in kg CO_2 per euro, and SEK import values are converted at annual Eurostat reference rates.

The earlier literature uses WIOD or EORA and keys intensity on the importing firm's own industry. That construction gives every one of a firm's import lines the same intensity regardless of what the firm actually bought, so the product composition of imports never enters the measure at all. The source matters for the same reason: in EORA the median country-year contains eight industries, some spanning ten Swedish industry divisions, and country explains 63% of the variance in log intensity against 23% for industry. EXIOBASE also separates direct from embodied intensity, and the two correlate only 0.70 in logs.

The difference is large enough to decide whether an instrumental-variables design is feasible at all. Instrumented with the shift-share that Section 5 builds, the EORA firm-keyed measure yields a first-stage F of 0.96, against 6.68 for the underlying import volume on the same sample: the emission weighting, not the instrument, was destroying the first stage. Moving to product-keyed EXIOBASE embodied intensity restores it. Source and key were varied one at a time, so the difference is attributable to the change that produced it.

I also compute, for each firm-year, the counterfactual intensity of the same import basket had it been produced in Sweden, using Swedish EXIOBASE intensities for the same products. The ratio of the two is the object that makes offshoring an environmental question instead of an accounting one: in the sample described below the median imported basket is 1.64 times as emission-intensive as its Swedish counterpart, and the mean 2.18.

To ask whether offshoring tracks policy arbitrage I also use the OECD Environmental Policy Stringency index, a composite of roughly fifteen market and non-market instruments, available 2004–2014. I classify an origin as lax if its index lies below Sweden's in the same year. Appendix B describes the index and the classification.

Emissions embodied in production are not the whole footprint of an import. Moving goods emits as well, and a firm that switches from a domestic to a distant supplier adds transport emissions that no production-side measure captures. I assign each import line a distance from Sweden to the origin country and a modal emission factor, and aggregate to firm-year freight CO_2; Appendix O gives the construction. Freight is small relative to embodied production emissions (a median of 2.4% of imported carbon, a mean of 5.0%), but it enters the decomposition in Section 6.4 as one of the five margins, and its sign matters because it works against the firm, not for it. Mode is inferred from geography, never observed, and import weights are missing for

part of the sample, so the series is a lower bound: where I cannot assign a mode I assign sea, the least carbon-intensive of the four. Appendix O gives the construction and plots the level and the share over the window.

3.3 Sample and summary statistics

Table 1 traces the sample from the universe of Swedish firm-years to the estimating sample. Two restrictions do nearly all the work. Requiring manufacturing takes 5.1 million firm-years to 366,354. Requiring reported energy use, the binding constraint since SCB surveys only plants with ten or more employees, takes that to 29,738, or 8.1% of manufacturing firm-years. Requiring positive value added and intermediate imports leaves 18,535, and a non-missing instrument 15,763. Matching freight gives the estimating sample of 15,749 firm-years and 3,181 firms.

The sample is 4.3% of manufacturing firm-years and 83.4% of the CO_2 reported by manufacturing. That gap is the point: emissions in manufacturing are concentrated in large plants, which are precisely the plants SCB surveys and precisely the plants that import. A sample covering a small share of firms and most of the emissions is the normal case in this literature, and it means the estimates speak to the bulk of the manufacturing carbon footprint even though they do not speak to the median Swedish manufacturer.

Table A.3 in the appendix compares firms in the estimating sample with manufacturing firms outside it. The estimating sample is larger and more emission-intensive by construction, because it is importing manufacturers. That is a statement about scope, and every specification below is identified within firm.

Table 1: From the firm universe to the estimating sample

Restriction	Firm-years	Firms	% mfg	Mt CO_2	% mfg CO_2
All firm-years, 2005-2014	5,126,653	956,583	1399.4	114.68	116.2
manufacturing	366,354	56,863	100.0	98.69	100.0
+ emissions reported	29,738	5,138	8.1	98.69	100.0
+ value added positive	29,566	5,124	8.1	98.26	99.6
+ intermediate imports (BEC)	18,535	3,640	5.1	96.96	98.2
+ both EXIOBASE intensities	18,535	3,640	5.1	96.96	98.2
+ instrument non-missing	15,763	3,183	4.3	82.28	83.4
+ freight matched *(estimating sample)*	15,749	3,181	4.3	82.28	83.4

Notes. Restrictions are cumulative. Shares are expressed relative to manufacturing, not to the full firm universe, so the first row exceeds 100%. CO_2 is stationary combustion computed from reported fuel use; it is defined only for firms that report energy, so the CO_2 column is constant across the first two restrictions by construction and is a share of *reported* manufacturing emissions from the third row on. The estimating sample is 4.3% of manufacturing firm-years and 83.4% of reported manufacturing CO_2.

Table A.2 reports the estimating sample in levels. The median firm employs 47 people, sells

96 MSEK, and emits 76 tonnes of CO_2 from stationary combustion; the mean firm emits 5,225 tonnes. The distance between those two numbers, a factor of 68, is the concentration described above and the reason every specification below includes firm fixed effects and clusters at the firm.

Three features of the import data shape what follows. First, sourcing is narrow: the median firm imports 4 distinct EXIOBASE products from 5 distinct origins in its first sample year, with means of 5.1 and 7.0. Sourcing sets this small are what make the fixed-cost structure of Section 2 the right framework, and they are what make product-keyed intensity informative: with four products, which four matters. Second, imported baskets are dirtier than their Swedish counterparts, at a median intensity of 0.61 against 0.37 kg CO_2 per euro. Third, baskets change: the median firm turns over 4% of its basket relative to its own first year, but the mean is 26% and the 90th percentile is complete replacement. Most firms source stably, and a minority reallocate wholesale. That heterogeneity, not the average, is what the instrument exploits.

4 Carbon-Intensity Gaps and Sourcing Patterns

Section 3 described the sample. This section establishes the three facts the rest of the paper rests on: that the carbon intensity of production differs sharply across the countries Swedish firms buy from, that the gap holds industry by industry and is no artefact of national averages, and that it survives all the way down to the individual firm's own import basket.

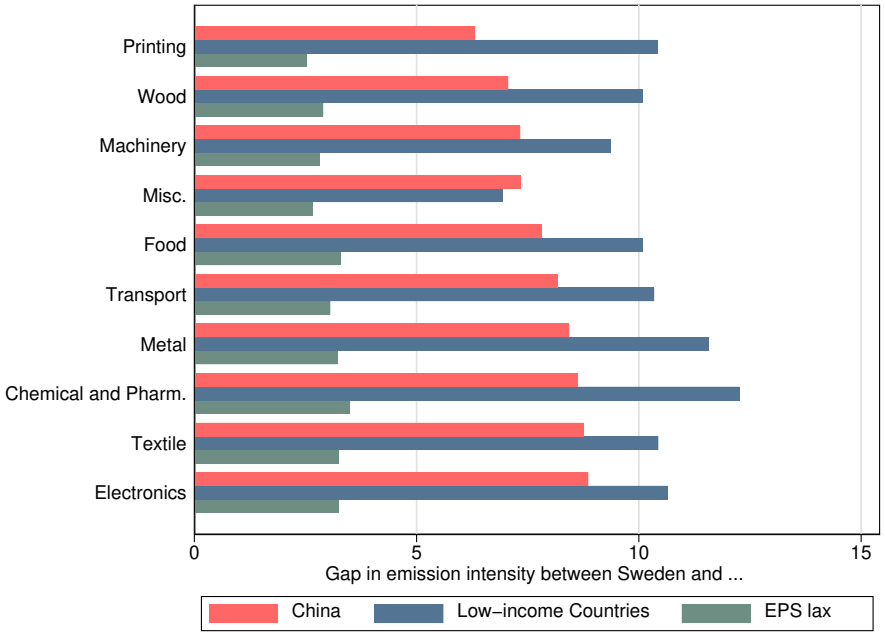
The firm-level fact is the one that matters for identification, so I state it first. For the median firm in the estimating sample, the goods it imports carry **1.64 times** the carbon that producing the same basket in Sweden would have released, at a median import intensity of 0.61 against a Swedish counterfactual of 0.37 kg CO_2 per euro (Table A.2). The mean ratio is 2.18, so the distribution is skewed and a minority of firms buy baskets several times dirtier than the domestic alternative. Reallocating a given basket toward these origins therefore raises embodied carbon even when the firm buys exactly the same products in exactly the same quantities. That is the margin the instrument is built to move.

Where the gap comes from. The policy-stringency split is descriptive and the identification does not rest on it. Splitting the instrument by origin group leaves no first stage, so the paper cannot say whether the gap reflects environmental policy, the energy mix, or the sectoral composition of what each country exports. Appendix B plots emission intensity by income group and by policy stringency: Sweden sits far below both comparison groups throughout the period and the gap does not close. What the argument needs is weaker and better identified. The gap

is large, it is sectoral, and it is priced into the basket a firm buys, so reallocating a given basket toward these origins raises embodied carbon with no change in what is bought.

The gap holds industry by industry, and Figure 2 shows its size industry by industry. Sweden is roughly eight times more carbon efficient than China, ten times more efficient than the average low-income country, and four times more efficient than countries whose environmental policy is laxer than its own. The largest gaps are in metals, transport and chemicals, the energy-intensive industries; but textiles and electronics, which are not energy-intensive, show large gaps too. That pattern is itself informative: if the gaps tracked environmental policy, they would not appear where energy is a small share of costs. What the figure establishes is that reallocating a given basket toward these origins raises its embodied carbon substantially, whatever the reason the gaps exist.

Figure 2: Sectorial Gap in emission intensities



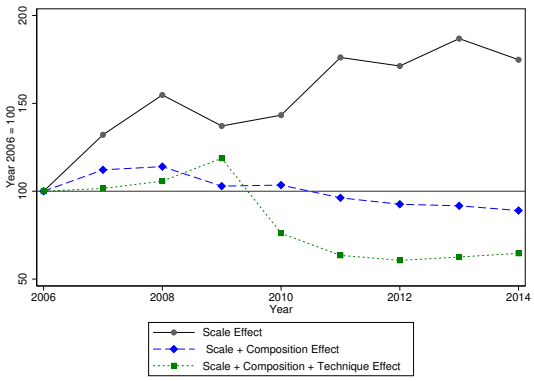
Scale, composition and technique. Before turning to firms, it is worth asking what the aggregate record can and cannot establish. Following Choi et al. (2023), Copeland et al. (2022) and Ustyuzhanina (2022), I decompose Swedish manufacturing CO_2 into the scale of the sector, the mix of industries within it, and the emission intensity of each industry. Appendix C gives the construction.

Figure 3 reports it, indexing aggregate manufacturing CO_2 to 1 in 2006. Scale works against the decline, since output grew over the period, and composition works against it too: the sector

moved toward more pollution-intensive goods, which is what Ustyuzhanina (2022) also report for Swedish manufacturing over 2007–2017. Technique had to offset both.

The decomposition cannot go further than that, and the reason is why this paper exists. The composition term counts any shift in the industry mix, whatever produced it, so a shift towards cleaner domestic production is observationally identical here to one that moves dirty production abroad. No decomposition built on emissions released inside Sweden can separate them. The firm-level analysis below can.

Figure 3: Decomposition of aggregate manufacturing emissions



Notes. Aggregate Swedish manufacturing CO_2 from equation (10), indexed to 1 in 2006. The black line is scale, the gap to the blue line is composition, and the gap to the green line is technique. The composition term counts any shift in the industry mix, whatever produced it.

Overall, despite a substantial increase in manufacturing output measured in monetary value, air pollution emissions from Swedish manufacturing firms fell by 29%. In comparison with most other countries, and despite its limited size, Sweden has a long tradition of strict environmental policies. In 1991 a carbon tax was introduced as a complement to the existing system of energy taxes (Johansson, 2000; Jonsson et al., 2020). The system has changed several times, but a common feature is higher rates for large carbon-emitting industries, and the country now levies the highest carbon tax rate in the world, at SEK 1,409 (US$129.89) per metric tonne of CO_2.[3] These regulations have historically mattered for reducing emissions from both stationary and mobile sources. Although they may have moved some firms towards cleaner technologies, the measured reduction also reflects the composition of net trade flows across industries. The effect of the carbon tax on the energy and resource efficiency of Swedish manufacturing can therefore be limited, once offshoring becomes an alternative to abating at home.

Concentration of Carbon offshoring. To give a rich background on the historical trends and changes in offshoring, I show in Figure P.1 the concentration of carbon offshoring within 2-digit

[3]See the World Bank carbon pricing dashboard.

industries between 1997 and 2014. I use these periods because the trade data go back to 1997, and I have enough sample years to create pre-sample product shares for my empirical purpose. The share of all firms engaged in offshoring, weighted by labour shares and sectors' labour weights, declined from 27.5% in 1997 to 23.0% in 2005. However, this share gradually increased from 2006 (the beginning of my sample period) and reached 27.0% by 2014. Offshoring tends to be concentrated among a select few firms. On average, the five largest offshoring firms within each industry account for 76.6% of the total offshoring volume. Moreover, the concentration becomes even more pronounced when considering local markets. Specifically, the two largest offshoring firms within each industry and county contribute, on average, 87.7% of the total offshoring volume. These descriptive findings highlight the dynamic relationship between offshoring activities, firm emissions, and the importance of properly controlling for firm characteristics in the regression analysis, as firms may self-select into offshoring. This also suggests that after 2005, the offshoring trends have been stable within sectors which is crucial for the construction of my instrument.

5 Empirical Strategy

A firm that buys more carbon-intensive inputs is rarely doing only that. It may be importing more because it is growing, in which case emissions and imports rise together for a reason that has nothing to do with sourcing; or it may be reorganising production and replacing a domestic step with a foreign one, which is the thing I want to measure. The raw correlation cannot tell these apart.

I address this problem by instrumenting firms' embodied-carbon imports with foreign export-supply movements, weighted by firms' pre-existing product–country sourcing relationships. The design uses differences in firms' initial exposure to foreign suppliers and subsequent changes in those suppliers' exports to markets outside Sweden and its close trading neighbours. The identifying assumption is that, conditional on the fixed effects, these movements affect Swedish firms' domestic emission intensity through their effect on embodied-carbon imports alone. I examine this assumption below, including a placebo test designed to detect common product-level shocks.

5.1 Estimand and baseline specification

I estimate the effect of embodied carbon in intermediate imports on firms' domestic emission intensity using

$$\ln\left(\frac{E_{it}}{VA_{it}}\right) = \alpha \ln CarbImp_{it} + \theta_i + \theta_{kt} + \theta_{mt} + \varepsilon_{it}, \tag{6}$$

where E_{it}/VA_{it} is firm i's stationary-combustion CO_2 emissions per unit of value added in year t, and $CarbImp_{it}$ is the carbon embodied in its intermediate imports, constructed as described in Section 3. The specification includes firm fixed effects θ_i, industry-by-year fixed effects θ_{kt}, and municipality-by-year fixed effects θ_{mt}. Standard errors are clustered by firm.

The coefficient α is the elasticity of domestic emission intensity with respect to embodied-carbon imports, and it is worth being precise about what that does and does not mean. It is not the elasticity of absolute domestic emissions. It is not the change in global emissions either. Emission intensity is a ratio, so a fall in it can come from the numerator or from the denominator, and I keep those two possibilities separate throughout.

Each set of effects does a different job. The firm effects take out whatever is fixed about a firm across the decade, which includes where it sits, what it specialises in, and the vintage of its technology. Industry-by-year effects remove anything common to an industry in a given year, so that a sector-wide demand movement or a change in industry regulation cannot by itself drive the estimate. Municipality-by-year effects do the same locally, and they matter here because local conditions and energy infrastructure differ across Sweden and move over time.

I report specifications that add these effects one rung at a time, and I treat the one carrying all three as preferred. That choice does not rest on the count of controls. It rests on the placebo evidence set out below, which is what tells me the municipality-by-year effects are removing confounding variation, not the variation the instrument needs. Comparing across the rungs is therefore informative about how sensitive the estimate is to what remains unobserved.

OLS does not necessarily identify α. Firms' import decisions and domestic emissions respond to common changes in production and demand, while sourcing and abatement decisions may be jointly determined; and embodied carbon is measured using source-country emission intensities that are more aggregated than the underlying product-level trade data. The second and third of these attenuate the OLS coefficient toward zero. The first is ambiguous on a ratio outcome, since a demand shock that raises emissions and value added together need not move their ratio at all. I therefore instrument embodied-carbon imports with foreign export-supply movements weighted by firms' pre-existing sourcing patterns. The identifying variation, and the assumptions required for its causal interpretation, are described below.

5.2 Instrument construction

The instrument combines two sources of variation: firms' pre-existing product–country import shares and subsequent movements in foreign export supply. Let $\bar{\omega}_{ipc}$ denote the share of product p imported from country c in firm i's intermediate imports in 2005. Let WX_{pct} denote exports of product p from country c to the world in year t, excluding exports to Sweden and its close trading neighbours: Denmark, Norway, Finland, and Germany. Let $\overline{EI}_{pc}$ denote the corresponding base-period EXIOBASE emission intensity. The instrument is

$$Shock_{it} = \sum_{p} \sum_{c} \bar{\omega}_{ipc} \, \overline{EI}_{pc} \, WX_{pct}, \qquad Z_{it} = \ln(Shock_{it}). \tag{7}$$

The shares fix what a firm was exposed to before the sample opens: which products it bought, and from where. Movements in world export supply then change the conditions facing those particular relationships. Two firms in the same industry and the same municipality, facing identical local conditions, are therefore exposed to different shocks, because in 2005 they were buying different things from different places.

I weight the shares by emission intensity so that the instrument predicts the right object. Left unweighted, a shift-share built on export supply predicts how much a firm imports, when what I need predicted is how much carbon comes with it; and those are not the same thing, because a rise in exports from a clean origin raises volume without raising embodied carbon in proportion. Weighting by base-period intensity closes that gap.

I hold those intensities at their base-period values, so that a contemporaneous change in a source country's measured carbon intensity cannot enter the instrument through the weights. That is a narrow fix and I do not want to oversell it. It rules out one channel and leaves another open, since foreign export movements may still carry product-level shocks that reach Swedish firms directly. I turn to that threat, and to the evidence on it, below.

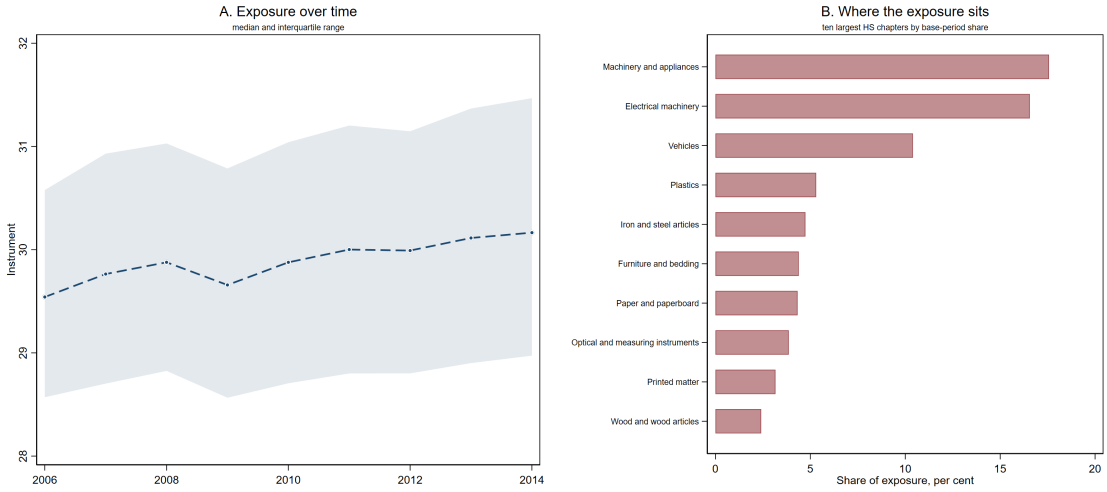
Excluding exports to Sweden and its close trading neighbours reduces the possibility that the instrument reflects Swedish demand for the imported products. If Swedish demand increased, exports directed to Sweden could rise alongside Swedish production and emissions. Removing these destinations makes the instrument less directly responsive to that demand. The exclusion does not, on its own, establish that the remaining export movements are exogenous: foreign supply, technology, transport costs, and global demand may still move together.

The construction therefore relies on a specific source of identifying variation. A shift-share instrument can be defended in either of two ways (Goldsmith-Pinkham et al., 2020; Borusyak et al., 2022): by assuming the exposure shares are exogenous, or by treating the shocks as the

random object and assuming they are as good as randomly assigned conditional on the fixed effects. I take the second of these, as Akerman et al. (2024) and Barrows and Ollivier (2021) do in the same setting. The shares here are each firm's pre-sample import composition, which I do not claim is exogenous: a firm that was already buying from dirty origins chose to. What the design asks is that movements in world export supply, specific to a product and an origin, are unrelated to whatever else would move a Swedish firm's emission intensity in that year. The shares enter as weights that determine which movements a given firm is exposed to, and because they are fixed before the sample and sum to one within each firm, the firm and sector–year effects absorb the level of exposure and leave its year-to-year movement to do the work. The next subsection examines the main threat to that assumption.

Figure 4 describes that variation. Exposure rises gradually across the window and is concentrated in a small number of product groups: machinery, electrical machinery and vehicles together carry 44.5 per cent of base-period exposure. Most of what the figure displays is absorbed before estimation. Of the instrument's variance, 0.8 percent survives firm, sector–year and municipality–year effects, and it is that residual variation, not the level or the trend, that identifies the estimates below. The fixed effects are demanding by design, and the first stage reported in Table 2 is computed on what they leave.

Figure 4: The instrument



Notes. Panel A plots the median firm-level instrument by year with the interquartile range; the median and not the mean, because a shift-share mean is dragged by the chapters Panel B identifies. Panel B gives the ten Harmonized System chapters carrying the largest share of base-period exposure, where a chapter's share is the sum of firms' pre-sample product shares falling in it; machinery, electrical machinery and vehicles together carry 44.5 percent. The level and time path in Panel A are largely absorbed by the fixed effects and are shown to describe the instrument, not the variation used: 0.8 percent of the instrument's variance survives firm, sector–year and municipality–year effects, and that residual variation is what identifies the estimates in Table 2. *Sources:* UN Comtrade and EXIOBASE, as in Section 3.

5.3 Identification and threats to validity

The shift-share design does not require firms' initial import shares to be randomly assigned. Firms that imported steel from China in 2005 may differ systematically from firms that imported textiles from Portugal. Instead, the design relies on the assumption that subsequent movements in foreign export supply, interacted with these predetermined shares, are conditionally unrelated to shocks to Swedish firms' emission intensity.

The firm effects absorb whatever is persistent about a basket, and the industry-by-year and municipality-by-year effects take out shocks shared at those levels. What identifies the estimate is then narrow: differences across firms in how exposed they were, in 2005, to the particular products and origins whose world supply later moved. One concern survives all of this. A movement in foreign export supply may coincide with a product-level shock that reaches Swedish production whether or not a firm imports anything.

Take steel. A process innovation in steelmaking raises exports from steel-producing countries, and it may at the same time change the production costs or the emission intensity of the Swedish firms that use steel. Firms more exposed to those exporters in 2005 would then show different domestic emissions for a reason that has nothing to do with how much carbon they imported, and my instrument would pick it up. Industry-by-year effects are no guarantee against this, because two firms in the same industry can buy different products from different places.

To see whether that is happening I build a placebo. The outcome is the emission intensity of Swedish production of exactly the products each firm imports: it moves when the domestic technology for those products moves, and it cannot respond to any single firm's sourcing decision, which is what makes it useful here. If the instrument predicts it, the instrument is reaching Swedish production without passing through imports at all. That would be fatal.

It does, at the looser rungs. With firm and industry-by-year effects alone the instrument predicts the placebo, and that is a failure. Adding municipality-by-year effects, the relationship is no longer distinguishable from zero. This is why I prefer the specification with all three sets, and the reason is the placebo, not the control count. I do not claim the test rules out every direct effect foreign export supply might have; what it shows is that the looser specifications are exposed to confounding the preferred one removes.

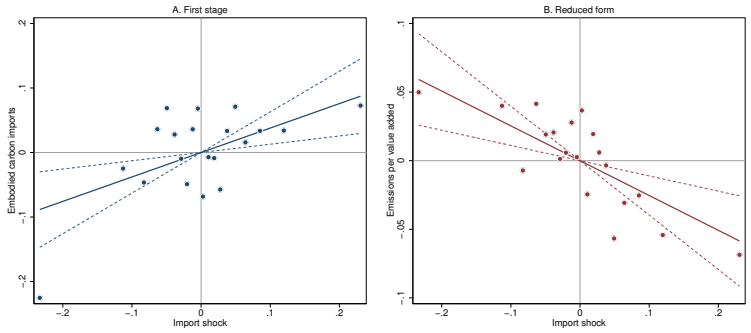
Adding the firm's effective carbon price as a control leaves the coefficient where it was. I read that narrowly. It says the estimate is not being driven by differences in what firms pay for carbon, and it says nothing more, because a carbon-price control reaches one channel out of many and the control is itself plausibly a function of the sourcing and production choices I am trying to explain.

Taken together, the fixed effects, the destination exclusions, and the placebo test support the interpretation of the instrument as a source of variation in embodied-carbon imports. The causal interpretation nevertheless rests on the conditional exclusion restriction that the instrument affects domestic emission intensity through embodied-carbon imports and through no unobserved product-level channel.

5.4 Estimation and scope

Figure 5 presents the first stage and reduced form after absorbing the fixed effects in the preferred specification. The first stage relates the instrument to embodied-carbon imports; the reduced form relates the same instrument to domestic emission intensity. Reporting both is useful because the first stage shows whether the instrument predicts the endogenous regressor, while the reduced form shows whether the instrument predicts the outcome without dividing by the first-stage coefficient.

Figure 5: First stage and reduced form



Notes. Binned scatters after absorbing firm, sector-year and municipality-year effects, so each slope is the corresponding coefficient. **Left:** the shock against embodied carbon imports, the first stage. **Right:** the shock against emission intensity, the reduced form. The 2SLS estimate in Table 2 is the ratio of the two slopes.

The first-stage slope is 0.379 on 14825 firm-years, and the reduced-form slope is -0.254. Both sides of each regression are in logs, so the first-stage coefficient is the elasticity of embodied-carbon imports with respect to the instrument, and the ratio of the two slopes is the two-stage least squares estimate reported in Table 2.

Dividing one by the other gives the two-stage least squares estimate, which therefore inherits both the relevance of the instrument and the exclusion restriction set out above. I report the Kleibergen–Paap statistic as the conventional diagnostic for relevance. It is a diagnostic and not a clearance.

So I also invert the Anderson–Rubin test over candidate values of α and report the resulting

confidence sets, whose coverage holds however weak the instrument turns out to be, conditional as always on the exclusion restriction itself. The sets and their interpretation are reported in Appendix J. The comparison with OLS is informative if the confidence set excludes the OLS estimate, but that comparison does not by itself establish that the difference is caused by any single source of OLS bias.

The instrument is constructed from firms' pre-sample import shares. The resulting estimate therefore describes the response of firms that were already importing intermediate inputs and whose embodied-carbon imports respond to the instrument. It does not identify the effect of inducing a firm with no pre-sample imports to begin importing. That extensive-margin question requires a different source of identifying variation.

The sample is also not the population. It is a small share of Swedish manufacturing firms and a large share of Swedish manufacturing emissions, which is the normal shape of this kind of data and is why the estimates speak to established importers in the emissions-intensive part of the sector, not to the average manufacturer. I keep that scope in view when the estimates are carried to the aggregate.

The main results should be interpreted in light of these diagnostics and the identifying assumptions. The following section reports the estimated effect of embodied-carbon imports on domestic emission intensity and then examines the margins through which that relationship arises.

6 The Measured Clean-up

6.1 Embodied-carbon imports and domestic emission intensity

In Table 2 I report the effect of embodied-carbon imports on firms' domestic emission intensity. Column (1) is ordinary least squares. Columns (2)–(4) instrument embodied-carbon imports with the shift-share built from firms' pre-sample sourcing patterns and subsequent movements in foreign export supply, adding fixed effects as they go; column (3), with firm, sector–year and municipality–year effects, is the rung I prefer. Columns (5) and (6) keep everything and swap the regressor for the value of the same imports, which is the comparison Section 6.1 returns to.

Under the identifying assumptions set out in Section 5, a 10 percent increase in the carbon embodied in a firm's intermediate imports lowers its stationary-combustion CO_2 per unit of value added by about 6.7 percent. The elasticity is -0.670. What it does not say depends on where the emissions are measured. The outcome counts fuel burned at the firm's own site, so this is a decline in production-based intensity and not, by itself, a decline in the firm's absolute

Table 2: Offshoring and firm-level emission intensity

Instrument	–	Carbon-weighted				Unweighted
	OLS	IV	IV	IV	IV	IV
	(1)	(2)	(3)	(4)	(5)	(6)
Carbon import	-0.025***	-0.628**	-0.670**	-0.657**		
	(0.006)	(0.307)	(0.281)	(0.280)		
Carbon tax rate				-1.137*		
				(0.658)		
Intermediate Import					-0.785**	-0.343**
					(0.350)	(0.175)
Control	No	No	No	Yes	No	No
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes
Sector-Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Municipal-Year FE	No	No	Yes	Yes	Yes	Yes
Observations	14,825	14,825	14,825	14,825	14,825	14,821
KP		12.20	15.97	15.72	14.87	25.15

Notes. The sample is Swedish manufacturing firms importing intermediate inputs, 2006–2014. The dependent variable in all columns is log CO_2 over value added, where CO_2 is the firm's own stationary combustion. Column (1) reports OLS and columns (2)–(6) report 2SLS. Column (3) is the preferred specification; column (4) adds the firm's effective carbon tax rate. The regressor is log embodied carbon imports in columns (1)–(4) and the log value of imported intermediate inputs in columns (5) and (6), which are estimated on the column (3) sample. The instrument is a shift-share of world export supply on pre-sample product shares. Carbon-weighted means the product shares are weighted by the emission intensity of the exporting industry–country; column (6) leaves the same shift-share unweighted, which is the specification estimated by Akerman et al. (2024). The reported *F*-statistic is the first-stage Kleibergen–Paap *F*-statistic. Standard errors clustered by firm are in parentheses. *** 1%, ** 5%, * 10%.

emissions or in the global emissions behind what it makes and what it buys.

The estimate is similar across specifications. With firm and sector–year fixed effects, the coefficient is -0.628; adding municipality–year effects gives -0.670. Including the firm's effective carbon tax rate yields -0.657. This stability suggests that the preferred result is not driven solely by the local shocks absorbed by municipality–year effects or by the variation captured by the additional tax-rate control. Because the effective tax rate may itself respond to a firm's fuel choices, however, column (4) is best interpreted as a sensitivity check, not as the preferred causal specification.

The OLS estimate is much closer to zero, at -0.025. The difference between OLS and IV is consistent with the endogeneity and measurement problems discussed in Section 5; it does not, by itself, identify which source of bias accounts for the gap. I therefore assess the IV estimate using the first-stage and reduced-form relationships and report weak-instrument-robust Anderson–Rubin confidence sets in Appendix J. These sets provide inference that does not rely on a strong first stage, conditional on the validity of the instrument's exclusion restriction.

The central finding is that firms' domestic emission intensity responds negatively to an instrument-induced increase in the carbon embodied in their intermediate imports.

Columns (5) and (6) replace the regressor with the log value of the same imports, holding the sample and the fixed effects at column (3), and differ from one another only in the instrument. Column (6) uses the unweighted shift-share, which is the specification Akerman et al. (2024) estimate on Swedish manufacturing over 2004–2016 with the same outcome and the same family of instrument. They report an elasticity of -0.56 and I estimate -0.343. The two samples are not nested, so I do not test equality across them; each estimate lies inside the other's confidence interval.

Two measurement choices separate column (6) from column (3), and the columns change them one at a time. Replacing the unweighted instrument with the emission-weighted one moves the estimate from -0.343 to -0.785; replacing import value with embodied carbon then moves it to -0.670. The second step is the smaller of the two, and the first stages say why. Columns (3) and (5) share an outcome, an instrument and a sample, so they share a reduced form, and the ratio of their coefficients is the ratio of their first stages: the shock moves the value of a firm's imports by about six-sevenths of what it moves their carbon content, and two regressors that the instrument moves at nearly the same rate deliver nearly the same elasticity. Whether the first step is larger than the second in a statistical sense is a separate question, and one these columns do not settle.

The carbon weighting is needed for the accounting, not for the estimation. Sections 7.4 and N ask where the carbon goes and how much of the aggregate decline it accounts for, and import value cannot answer either question, because carbon is the object being counted.

6.2 The firm-level response and aggregate manufacturing emissions

A negative firm-level elasticity does not imply that offshoring explains the aggregate decline in Swedish manufacturing emissions. The elasticity describes how a firm's domestic emission intensity responds to a change in embodied-carbon imports. The aggregate contribution also depends on how embodied-carbon imports changed over time and on how firms' output and economic activity evolved.

Between 2006 and 2014, Swedish manufacturing emission intensity fell by approximately 94 log points. The paper's aggregate decomposition attributes most of this decline to changes within continuing firms, not to entry, exit or the reallocation of activity between firms. Applying the estimated elasticity to the observed change in embodied-carbon imports gives an estimated contribution of 18.6 log points in the opposite direction to the aggregate decline. On this accounting,

changes in embodied-carbon imports did not drive the aggregate clean-up over the period.

The distinction between import value and embodied carbon helps explain why this result differs from an accounting exercise based on import growth alone. Import values rose over the period, while the carbon embodied in those purchases evolved differently. The environmental implications of sourcing therefore cannot be inferred from import growth without also accounting for the carbon intensity of the imported basket.

This aggregate calculation is an accounting application of the estimated elasticity, not a separate experimental estimate of the contribution of offshoring to national emissions. Its interpretation depends on applying a firm-level response to the observed changes in embodied-carbon imports and on the assumptions used to aggregate across firms. It should consequently be read as evidence against attributing the observed aggregate clean-up to this particular margin of offshoring, and not as a complete decomposition of the causes of Sweden's emissions decline. Appendix N reports the decomposition and the assumptions required to carry a within sector–year estimate to an aggregate.

Taken together, the results establish two distinctions. First, the carbon content of imported inputs is more directly relevant to the environmental question than import expenditure alone. Second, a decline in domestic emission intensity at the firm level need not imply that offshoring accounts for the aggregate decline in manufacturing emissions. The following robustness analysis examines whether the firm-level estimate is sensitive to alternative samples, instrument definitions and specifications.

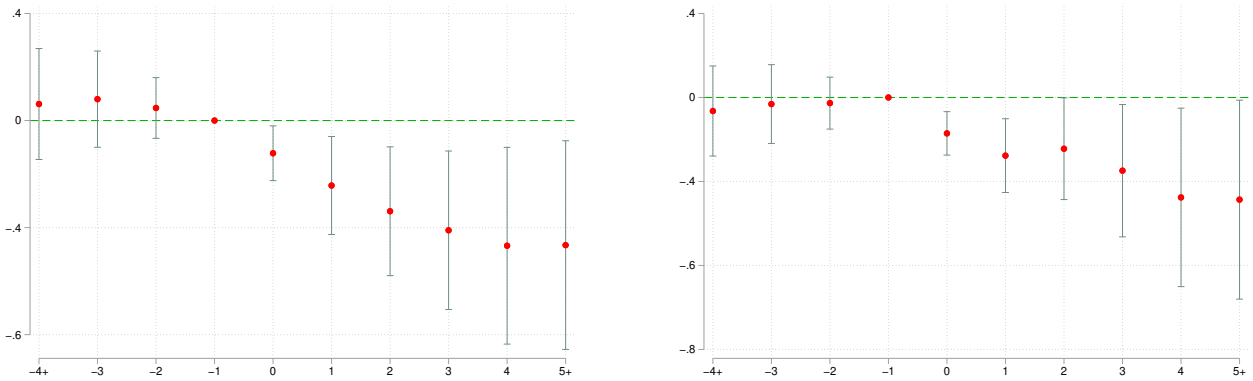
6.3 Robustness

Two things could make the estimate in Table 2 less than it appears. The first is that it rests on a particular set of sample and specification choices, so that different choices would give a different number. The second is that the instrument tracks a decline already under way, in which case the timing of the shock would be incidental, not causal.

In Figure K.1 I re-estimate the headline specification under a series of perturbations: dropping the financial-crisis years, adding policy controls, trimming the instrument at the first and ninety-ninth percentiles, winsorising the outcome, and clustering on firm and sector–year together. None is decisive on its own. The estimate moves within its confidence interval and does not change sign. A perturbation that could plausibly matter is dropping firms covered by the EU Emissions Trading System, since those firms face a second carbon price the instrument does not capture. Appendix G estimates that split directly, alongside a split by how carbon-intensive the firm's sector is: firms outside the system give -0.677, against -0.670 in the pooled sample, and

neither difference is large enough for the data to detect. Appendix K varies the design choices behind the preferred column one at a time: the fixed-effect rung, the clustering, the baseline-scale control and the denominator.

Figure 6: The clean-up is dynamic, not a one-year blip



A. CO_2 per unit value added

B. Energy per unit value added

Notes. IV event-study coefficients β_m, $m = -4, \ldots, 5$, with $\beta_{-1} = 0$; 95% confidence intervals clustered by firm; balanced 2006–2014. Leads sit on zero and the decline begins at the shock and keeps growing. Carbon and energy intensity trace the same path, so the response is a change in how the firm runs, not a one-off accounting shift. Intervals widen at long horizons because fewer firms are observed four and five years out.

The second concern is harder, because it is about timing, not specification. A shift-share instrument built on world export supply would be compromised if the firms exposed to the largest shocks were already on a declining path when the shock arrived. In Figure 6 I therefore trace the response over event time, normalising the year before the shock to zero. The leads sit on zero, so the instrument is not tracking a decline already under way. The decline begins at the shock and continues to grow for four to five years afterwards, and the intervals widen at the longest horizons because fewer firms are observed that far forward.

Appendix I repeats the check without instrumenting anything, by building the event-time dummies from the timing of the shock instead of the firm's imports. That version does not rest on the first stage at all, its leads are jointly indistinguishable from zero, and its path after the shock has the same shape. Two ways of asking whether the timing is incidental give the same answer.

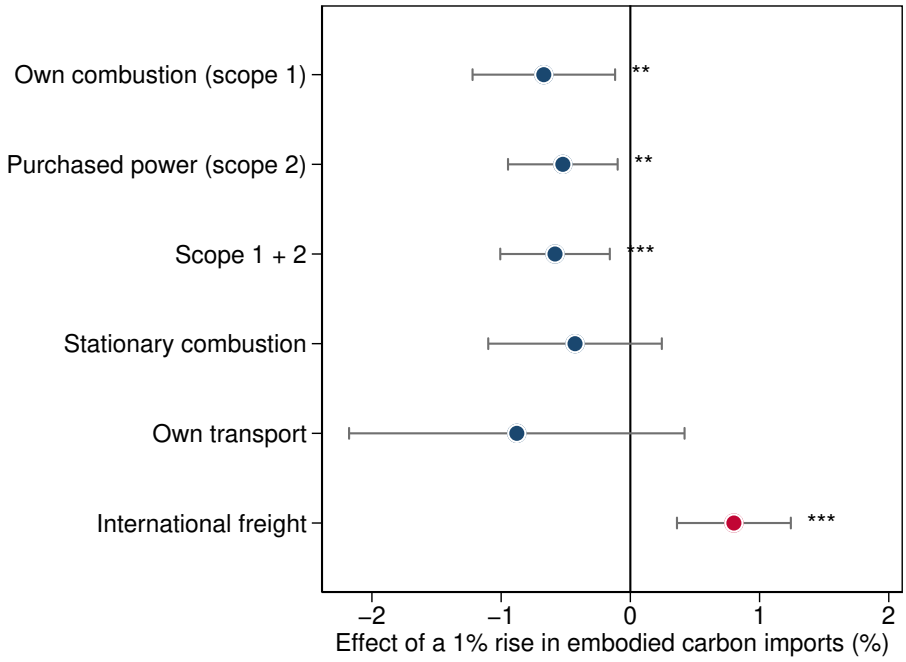
Carbon intensity and energy intensity trace the same path in the two panels. That matters for interpretation: if the response were an accounting artefact, there would be no reason for the physical energy series to move with it. A firm changing how much energy it uses per unit of output is a different object from a firm relabelling an emissions line, and the two panels together are what distinguish them.

6.4 Which emissions move

Emission intensity is an aggregate of several things a firm burns, divided by what it produces. An elasticity of -0.67 on that aggregate does not say where the fall sits. This section asks two questions of it: which emissions move, and whether what moves is the amount of energy a firm uses or the carbon content of that energy. The specification never changes: the same instrument, the same fixed effects and the same sample as column (3) of Table 2, with only the dependent variable moving. Why the firm uses less energy is the subject of Section 6.5, and what happens to the carbon that leaves its books is the subject of Section 7.

Figure 7 re-estimates the main equation with each component of the firm's carbon account on the left-hand side, each scaled by value added. The components are not independent draws, since scope 1 and scope 2 sum to scope 1+2 and stationary combustion and own transport partition scope 1, so the figure is an accounting statement. What it shows is which parts of the account move together.

Figure 7: Where the emission effect goes



Notes. Each row is a separate 2SLS regression of that component per unit of value added on log embodied carbon imports, instrumented as in Table 2. Bars are 95% confidence intervals. Components are nested, not additive: scope 1 is own combustion, the sum of stationary and transport fuel; scope 2 is purchased electricity and heat. **International freight is the only component that rises, and the only one outside the national inventory.** Emissions from moving goods to Sweden are attributed to bunkers and to the exporting territory, so a firm that offshores looks cleaner in the production-based account by construction. Firm, sector-year and municipality-year effects; standard errors clustered by firm. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Own transport deserves separate attention because the literature on trade and the environment has concentrated on emissions from production, and pollution from moving goods has received comparatively little of it (Copeland et al., 2022). That gap matters, since the damage from trade-related transport is exactly what policy debates about offshoring turn on. The firm's own transport fuel is a *lower bound* on that margin: it excludes the international shipping of the goods the firm imports, which appears on no manufacturer's books at all and which I take up below.

6.5 Energy use and the fuel mix

A firm can emit less per unit of output in two ways: by using less energy, or by using energy that carries less carbon. Carbon intensity is the product of the two, so $\ln(CO_2/VA) = \ln(E/VA) + \ln(CO_2/E)$, and the split is an identity. In panel A of Figure 8 I report both terms. Of the headline -0.67, energy per unit of output contributes -0.60 and the carbon content of that energy -0.06, and the second term is not significant. The decline is firms using less energy, not firms burning cleaner fuel. Appendix L repeats the split on the underlying energy and carbon quantities instead of the intensities; energy is what the survey measures and CO_2 is constructed from it by fuel-specific factors, which is why the two rows add exactly. Appendix M repeats the component decomposition at each of the three fixed-effect rungs, so that a conclusion appearing only at the loosest one can be told apart from one that survives all three.

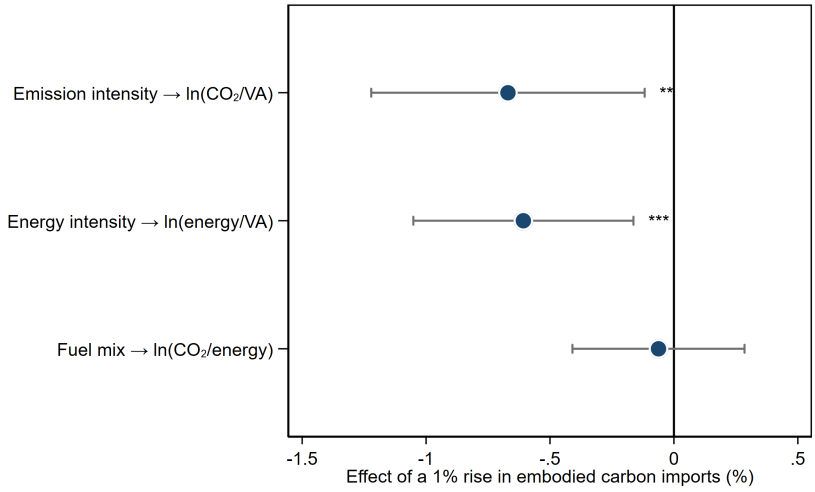
That answer raises the question Section 6.5 takes up. Nothing about buying dirtier inputs mechanically reduces the energy a firm needs at home, so the identity locates the response without explaining it.

Nothing about buying inputs that carry more embodied carbon mechanically reduces the energy a Swedish plant needs. Two explanations are available. The firm might be doing the same things with less energy, which is a technique effect and would show up as cleaner fuel or better efficiency at a given output. Or it might be doing different things, having moved the energy-intensive step of its own production abroad along with the input, which is a composition effect operating inside the firm. The two have different implications for whether the clean-up travels: a technique effect is a genuine improvement that would survive the trade channel closing, while a composition effect is a relocation that would not.

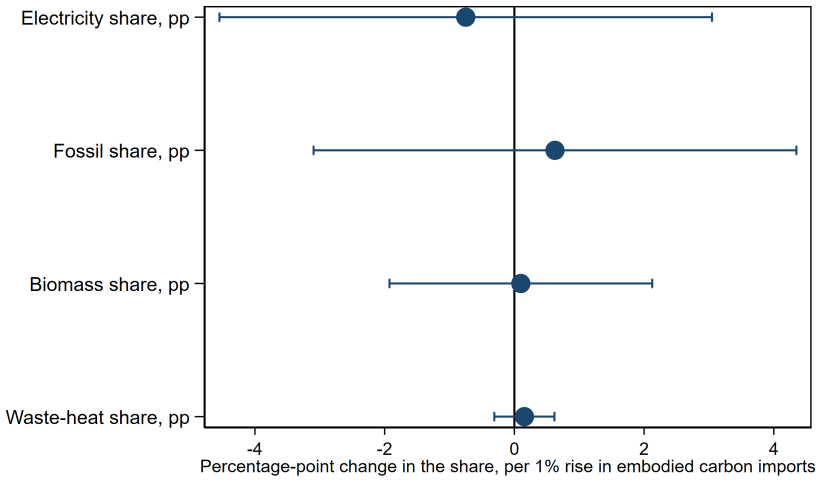
The fuel evidence rules out the first. Because $\ln(CO_2/E)$ enters the identity as a residual, an insignificant estimate on it is weak evidence; I therefore put the energy mix itself on the left-hand side, as shares of total gigajoules, and test it directly. Panel B of Figure 8 reports the result on the strongest first stage in the paper, a Kleibergen–Paap F of 22.4. No share moves.

Figure 8: Less energy, or cleaner energy?

A. The headline elasticity split into its two terms



B. The fuel shares themselves



Notes. Panel A splits the headline elasticity using $\ln(\mathrm{CO}_2/VA) = \ln(E/VA) + \ln(\mathrm{CO}_2/E)$, an identity, so the lower two rows sum to the top one. The decline is energy per unit of output, not a change in what the firm burns. Because the carbon-content term enters that identity as a residual, panel B tests it directly: shares of total GJ on the left-hand side, in percentage points, at a Kleibergen–Paap F of 22.39. No share moves. **These firms already draw 77 percent of their energy as electricity, so little switching margin remains.** Unlike $\ln(\mathrm{CO}_2/E)$, panel B does not condition on positive combustion CO_2, so firms that switched completely remain in the sample.

Electricity shifts by -0.75 percentage points, fossil fuels by $+0.63$ and emissions per gigajoule by $+1.36$, none of them distinguishable from zero against standard errors close to two percentage points. Appendix D reports the same exercise for abatement investment, where the instrument has almost no power on the survey subsample and no finding is available in either direction. The specification does not condition on positive combustion CO_2, so firms that abandoned fossil fuel entirely remain in the sample and a complete switch would be visible if it had occurred.

The reason no share moves is in the base, not in the coefficient. These firms already draw 77 percent of their energy as electricity. There is very little dirty fuel left for them to switch away from, which is a feature of Swedish manufacturing specifically and a limit on how far this result travels: in a country whose industrial energy is still largely fossil, the same shock could plausibly produce a fuel-switching response that Sweden cannot generate.

What the shock does move is the scale of domestic activity. Value added rises by 0.348 and sales by 0.305, both significant at the one percent level and both estimated on first stages above 13. Section 7.1 re-estimates them at the fixed effects of column (3) of Table 2, where value added rises 0.463 instead; that is the rung at which the carbon identity closes, and it is the one the decomposition uses. Appendix E gives every outcome estimated on this specification. Emissions in levels and measured total factor productivity are estimated imprecisely, at first-stage F of 8.6 and 9.7, and I treat neither as a null: an underpowered estimate is not evidence of no effect, and I do not claim either that emissions are flat in levels or that productivity is unchanged. Productivity in particular deserves more than one underpowered row, and Appendix E says what is needed to settle it.

Taken with the placebo, this points to composition and away from technique. The domestic-technique margin of Section 6.4 is insignificant, the fuel mix does not move, and output rises while energy per unit of output falls. A firm that had become more efficient at what it was already doing would show a technique response somewhere; one that had moved its most energy-intensive step offshore and expanded the rest would show exactly this pattern. What the data record is the energy signature of that step, not the step itself, which is as close to it as firm-level fuel use can bring the question.

7 The Limits of Production-Based Accounting

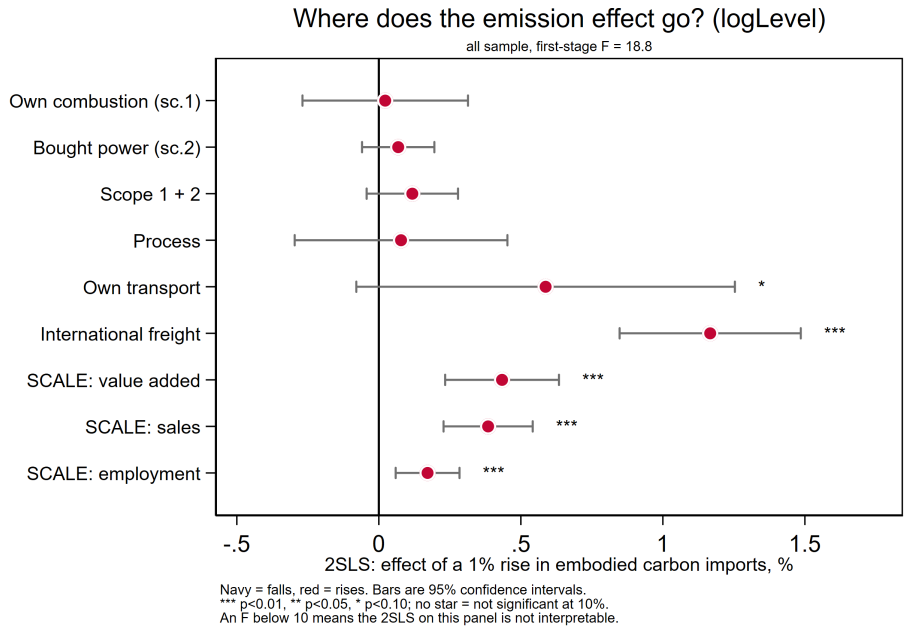
Every figure a country reports under its climate commitments is production-based: it counts what is emitted inside its own borders. Sweden's national inventory rests on that convention, so does the base of its carbon tax, so does the Emissions Trading System, and so does the firm-level literature that measures manufacturing clean-up. The convention has a seam. It does not count the fuel a firm burns moving goods itself, and it does not count the bunkers of the ship carrying its imports, which sit on no manufacturer's inventory in any country. Those two margins are where a firm's reported performance and its actual footprint come apart. The shock moves both.

7.1 Output or emissions?

An intensity can fall because its numerator shrinks or because its denominator grows, and only the first is a story about carbon. Removing the denominator settles which it is, and it uncovers something the ratio had concealed. In Figure 9 I report every component of the firm's account as a level, with no denominator anywhere.

Everything the national inventory counts is flat. On-site combustion, bought power, scope one and two together, process emissions: none of them moves. The only emissions that rise are the firm's own transport and the international freight carrying its imports, which are precisely the two the inventory does not record. The carbon the ledger records does not respond to the shock, and the carbon it ignores does.

Figure 9: The same components in levels

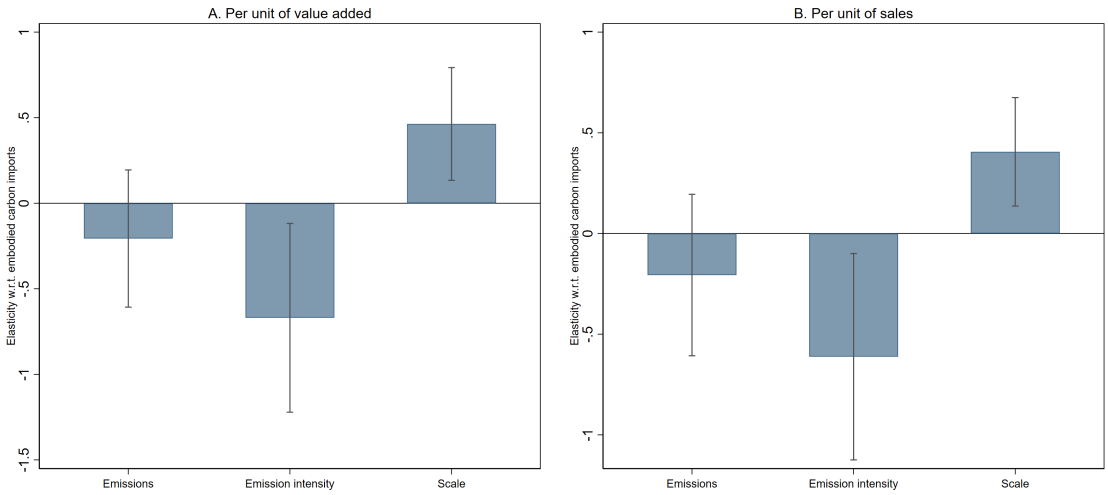


Notes. No denominator: every row is a level. **Everything the inventory counts is flat; the only emissions that rise are own transport and international freight.** Scale rises as well. Firm and year effects, full sample.

The arithmetic behind the exercise is immediate. Because $\ln(E/VA) = \ln E - \ln VA$, the elasticity of the intensity is the difference between the elasticity of emissions and that of value added, so each side can be estimated on its own. I scale by value added throughout, the denominator implied by the firm-level decomposition of Cherniwchan et al. (2017), because emissions per krona of revenue fall almost mechanically when a firm moves tasks off site.

Figure 10 closes the identity. Estimated on the 14825 firm-years where all three outcomes are defined, and at the fixed effects of column (3) of Table 2, emission intensity falls by -0.670, value

Figure 10: Does the firm get cleaner, or just bigger?



Notes. Each bar is the 2SLS elasticity of the stated outcome with respect to log embodied-carbon imports, on 14825 firm-years with firm, sector–year and municipality–year effects and standard errors clustered by firm, as in column (3) of Table 2, whose coefficient the emission-intensity bar reproduces. Because $\ln \mathrm{CO}_2 = \ln(\mathrm{CO}_2/VA) + \ln VA$ holds observation by observation and 2SLS is linear in the outcome, the emissions elasticity is the sum of the other two exactly; the adding-up is verified to 10^{-8} before the figure is drawn, which requires the sample on which all three are defined and so a smaller one than Figure 9, estimated at firm and year effects on the full sample. Panel A uses value added as the denominator and Panel B sales, so the emissions bar is common to both and only the split differs. Capped lines are 95 percent intervals, each from its own regression. The decomposition follows Barrows and Ollivier (2021).

added rises by 0.463, and emissions themselves move by -0.206, which no test separates from zero. The three sum exactly, by construction and in the estimates. Just over two thirds of the measured intensity decline is the denominator growing, not the numerator falling, and the same split against sales gives -0.612 and 0.406. The production-based statistic improves; the emissions behind it do not detectably fall.

The two sides of that identity are not equally well measured, and the difference matters for what can be claimed. Value added rises 0.463 and sales 0.406 on this specification, both precisely estimated, so part of the measured intensity gain is more output, not less carbon. The flat rows are a weaker object: they are insignificant in both directions, one specification giving $+0.11$ and another -0.205 at a first-stage F of 8.6. Those rows constrain the on-site response in neither direction, and I draw nothing from them. What the figure establishes is which components moved, and those are the uncounted ones.

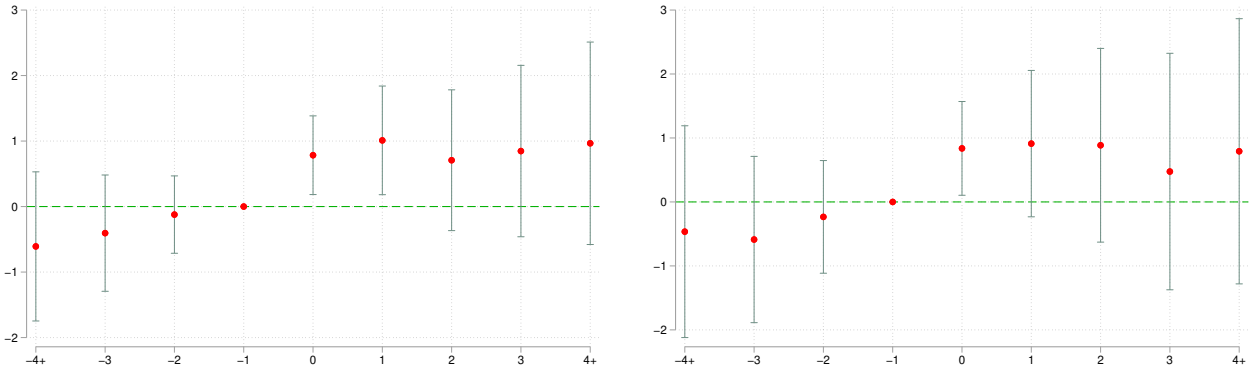
This locates the argument relative to Akerman et al. (2024), who use the same Swedish registers and attribute the import–emission link largely to cleaner technology. Whatever drives the domestic channel they identify, the masking documented here arises from emissions outside the production boundary, not from how on-site intensity is scaled. The two findings describe

different parts of the same account.

7.2 Own transport emissions

The firm's own transport fuel is the nearer half of the uncounted margin, and the only component of the account that responds positively to the shock. Figure 11 traces it over event time against the production-based series. Where measured intensity falls from the year of the shock onwards, transport emissions move in the opposite direction: the sign flips at the shock and the gap widens as the horizon extends. Both series come from one decision. A firm that replaces a domestic supplier with a distant one books a fall in its own combustion and an offsetting rise in the carbon of moving goods, and only the first appears in any inventory. Over the decade stationary combustion falls sharply while own transport is flat. And this is still only the firm's own lorries.

Figure 11: Own transport emissions over event time



A. Transport CO_2 **B.** Transport CO_2 per unit of value added

Notes. Instrumented event-study coefficients β_m with $\beta_{-1} = 0$, 95 percent confidence intervals clustered by firm, balanced 2006–2014. Panel A is transport CO_2 in levels; Panel B scales it by value added, so the reversal is not an artefact of output growing. In both, the sign flips at the shock and the on-impact and year-one effects are significant; from year two the estimates stay positive but the intervals are wide, so the figure establishes a sign reversal, not a precisely estimated long-run path.

The ships are the larger half, and they are not the firm's at all. International freight is carbon the domestic counterfactual never incurs, because goods made at home cross no border, and it is attributed to the carrier, not to the manufacturer or to either country. Emissions from moving goods have drawn far less attention than emissions from making them (Copeland et al., 2022), and the omission is not only academic: the European Union's carbon border adjustment prices the carbon embodied in an imported good and not the carbon spent bringing it, so the margin that grows fastest here is the one the instrument designed to stop leakage does not reach. I

construct it as

$$E^{\mathrm{FR}}_{it} \;=\; \sum_{c} M_{ict}\, d_c\, \varphi_{m(c)}, \tag{8}$$

combining the weight M_{ict} of firm i's imports from country c with the weighted distance d_c to that origin and a mode-specific emission factor $\varphi_{m(c)}$ in grams of CO_2 per tonne-kilometre, following Cristea et al. (2013). The modal factors come from the GLEC Framework v3 and the IMO Fourth Greenhouse Gas Study instead of that paper, and Appendix O gives the construction in full. Import weights are net mass, not gross, so E^{FR} is a conservative lower bound: a ship burns fuel moving the packaging and the vessel too.

7.3 The net footprint

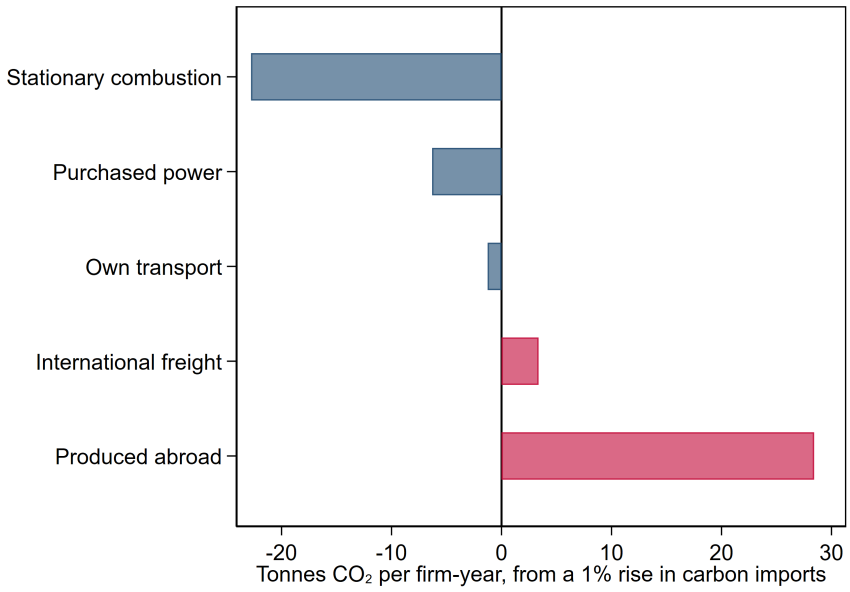
Adding the on-site, own-transport, freight and embodied-abroad components gives the firm's total carbon footprint, $E^{\mathrm{TOT}}_{it} = E^{\mathrm{DOM}}_{it} + E^{\mathrm{FR}}_{it} + E^{\mathrm{FGN}}_{it}$, the identity in equation (1). If E^{DOM} falls with offshoring while E^{FR} and E^{FGN} rise, the total need not fall at all, and a firm can clean up on the only ledger anyone reads while the carbon it is responsible for stays where it was. Freight is the sharper of the two uncounted components, because it is not a relocation at all. The carbon embodied in an imported input is carbon some producer would have emitted either way; the fuel burned carrying that input across an ocean is emitted only because the production moved. Offshoring does not merely shift emissions elsewhere. On this margin it creates emissions that no domestic arrangement would have produced, and attributes them to nobody.

Figure 12 puts the pieces in one unit. Each bar is the change in tonnes implied by the estimated elasticity for a firm at the sample mean, so the components can be added instead of compared as percentages. The first four bars are emissions the firm reports. The fifth is carbon embodied in what it imports, valued at the origin's emission intensity: carbon that is emitted, but not by the firm and not in Sweden.

That fifth bar is an accounting entry. The elasticity of foreign emissions with respect to carbon imports is one by construction, since the regressor and that outcome are the same variable, and I report it only because leaving it out would make the figure add up to a saving the accounting does not support. What the figure shows is the size of the domestic reduction set against the carbon that reappears abroad and in transit.

Two features of the construction bound how far the figure can be pushed. Its components come from registers with different coverage, so the bars are not all estimated on identical firm-years, which makes the figure a statement about magnitudes. And it holds each origin's emission intensity fixed, so it prices the carbon of today's foreign production technology, not what that

Figure 12: The components in tonnes



Notes. Each bar is the change in tonnes implied by the estimated elasticity for a firm at the sample mean, so the components add instead of being compared as percentages. The fifth bar, carbon embodied in imports, is the regressor itself expressed in tonnes: carbon embodied in imports (in kg) and the regressor are both built as import value times the origin's embodied intensity, so that bar and the total are close to arithmetic, not estimates. Read the first four bars, which are emissions the firm reports.

technology may become. Neither qualification touches the ordering the figure establishes, which is that the components moving upward are the ones no inventory records.

7.4 The margins of embodied carbon

A firm that offshores to a dirtier producer raises world emissions even as its own fall. That is the pollution-haven hypothesis, and the empirical literature has largely failed to find it. This section argues that the failure is partly a measurement problem and partly an averaging problem, and that neither is resolved by looking harder at the same margin.

The test the literature would run first is to split the instrument by the origin's environmental policy. That split has no first stage, for the reason Appendix F sets out: the same baseline product shares weight a firm's exposure to every origin group, so cutting the shock removes the identifying variation without delivering usable instruments. I therefore ask a different question of the same instrument.

Instead of asking which firms respond differently to a shock I cannot split, I ask which margin of carbon imports the pooled shock moves. Embodied carbon in a firm's imports is the product of three things: how much it buys, what it buys, and how dirty the origin is for the things it

buys. Writing that as an identity,

$$\ln C_{it}^{for} = \underbrace{\ln M_{it}}_{\text{volume}} + \underbrace{\ln\left(C_{it}^{swe}/M_{it}\right)}_{\text{product mix}} + \underbrace{\ln\left(C_{it}^{for}/C_{it}^{swe}\right)}_{\text{origin dirtiness}},$$

where C^{swe} values the same import basket at Swedish emission intensities, so the second term moves only with what the firm buys and the third only with where it buys it. The third term is the policy-arbitrage test.

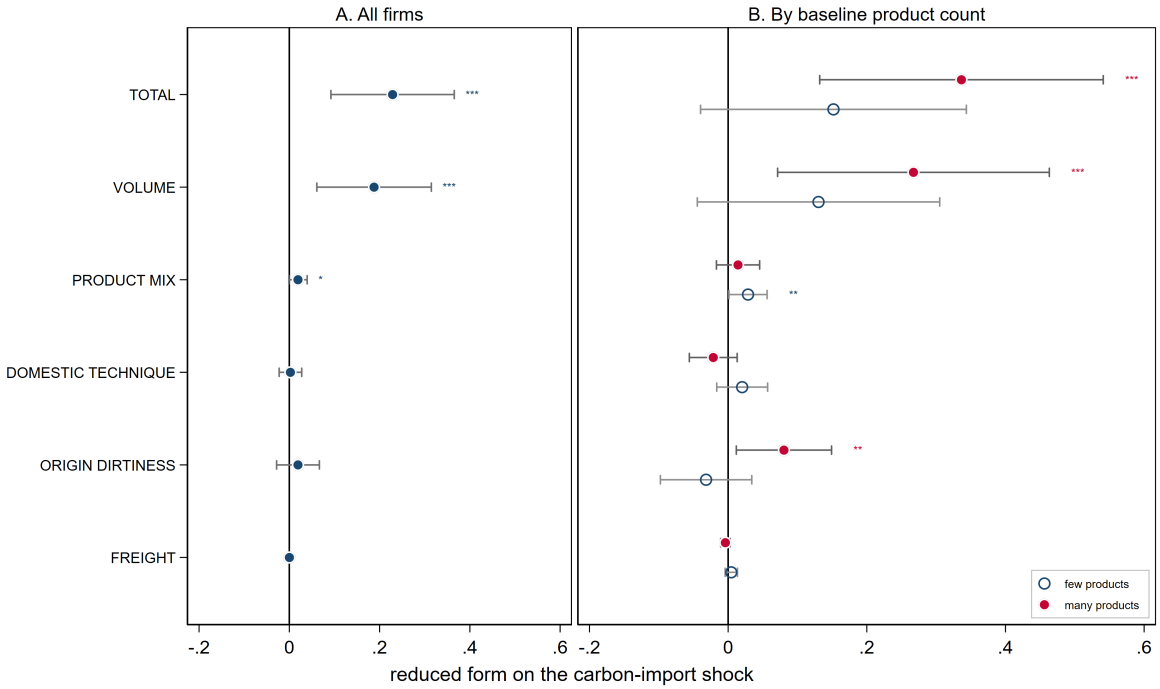
Each margin is regressed on the instrument directly instead of instrumented by it. That is not a weaker exercise here; it is the only coherent one. The margins are components of the endogenous variable itself, so instrumenting $\ln C^{for}$ with the shock and then decomposing it would amount to regressing the regressor on itself, which returns one by construction. The reduced form is the causal object: it answers what the shock does to each margin, the margins add up because the identity above is exact, and no first-stage strength is required for any of it. I add two further margins on the same basis. Domestic technique, which the shock should not move and which therefore serves as a placebo, and freight, which is emitted in transit and belongs to no producer's books. Prediction 2 signs that freight margin as a zero before it is estimated: the wedge between freight and the carbon it carries is invariant to a shock that does not favour distant origins over nearby ones, even while freight in levels rises, as Section 7 shows it does. Because the margins sum to the total by construction, nothing is learned from the total; the content is in how it splits.

In Panel A of Figure 13 I report the margins pooled over all firms. Volume carries four fifths of the total, which is Prediction 1: an origin-driven shock reaches the product-mix margin only to second order, so it should be an order of magnitude smaller than volume. It is. Origin dirtiness is not significant. That is the null the previous literature reports, and I reproduce it. The pooled estimate of origin dirtiness is not, however, smaller than the estimate of product mix. The two point estimates are the same and only their precision differs, so the pooled null is a statement about power, not about magnitude. A firm can only reallocate towards dirtier origins if it buys enough distinct things to have somewhere to reallocate, and the pooled average does not separate the firms that can from the firms that cannot. In Panel B I therefore split the margins on the number of distinct products a firm bought in its first sample year, fixed before any shock arrives.

This partition is the one Prediction 3 names, and the model's content is not that the origin margin is larger for broader firms but that the two composition margins should part company across the same cut. They do. Origin dirtiness is insignificant among firms that bought few

products and significant among those that bought many. Product mix, which the model treats as a technology object, is if anything larger among the narrow-scope firms. That is the opposite of what a story about firm capability would deliver: if broad firms simply respond more to everything, both composition margins would rise with scope, and so would volume. Domestic technique, the placebo, is insignificant in both groups. The pooled null averages over firms with no scope to substitute and firms with a good deal of it, and it is the second group in which the haven mechanism is visible. The average is the right summary of the population and the wrong summary of the mechanism.

Figure 13: The five margins of embodied carbon



Notes. Panel A pools all firms. Panel B splits them on the number of distinct products the firm bought in its first sample year, fixed before any shock arrives. Each bar is a reduced-form coefficient of the margin on the carbon-import shock, for the five margins of the identity in Section 7.4; they sum to the total by construction. Fixed effects as in column (3) of Table 2. Standard errors clustered by firm. Stars denote significance at 10, 5 and 1 percent.

7.5 Trade and ownership

The pollution-haven literature tests its hypothesis on foreign direct investment: firms build plants where regulation is lax and pollute there instead of at home. That is the natural test when the data are national aggregates, because ownership is what aggregate statistics record. With firm records the question can be put the other way round, by asking which sourcing relationship the

shock actually moves, and the answer is the arm's-length one.

Table 3 splits a firm's imports into what it buys from origins where it owns a plant and what it buys from everyone else, then runs the main specification on each. Arm's-length sourcing responds: the estimate is -0.676, and because its first stage is 8.61 the interval that matters is the Anderson–Rubin set, which runs from -1.93 to -0.32 and excludes zero. Coverage of that set does not depend on the strength of the first stage, so the claim survives the weakness of the design that produced it. The own-affiliate margin, by contrast, has a first stage of 0.16 and an Anderson–Rubin set that covers the whole searched grid. I report it to establish that the margin is uninformative, not that it is zero: the two rows are the same firms and the same instrument with a different regressor, two just-identified regressions, so they cannot be differenced.

Table 3: Multinational status and the sourcing margins

	N	OLS	2SLS	KP F	AR 95% set
Panel A. Groups and sourcing margins					
All firms	15,175	-0.025	-0.524***	13.92	[-1.12, -0.23]
			(0.183)		
MNE parents	322	-0.048	-0.249	6.86	[-1.85, 0.89]
			(0.464)		
Non-MNE	14,808	-0.025	-0.528***	13.20	[-1.16, -0.23]
			(0.188)		
Own-affiliate imports	11,587	0.001	4.379	0.16	$\supseteq [-10, 10]$
			(11.121)		
Arm's-length imports	11,587	-0.014	-0.676**	8.61	[-1.93, -0.32]
			(0.264)		
Panel B. Formal interaction, fixed effects crossed with MNE status					
Non-MNE			-0.528***	6.59	
			(0.188)		
MNE parents			-0.249	6.59	
			(0.456)		
Difference			0.279	6.59	
			(0.493)		

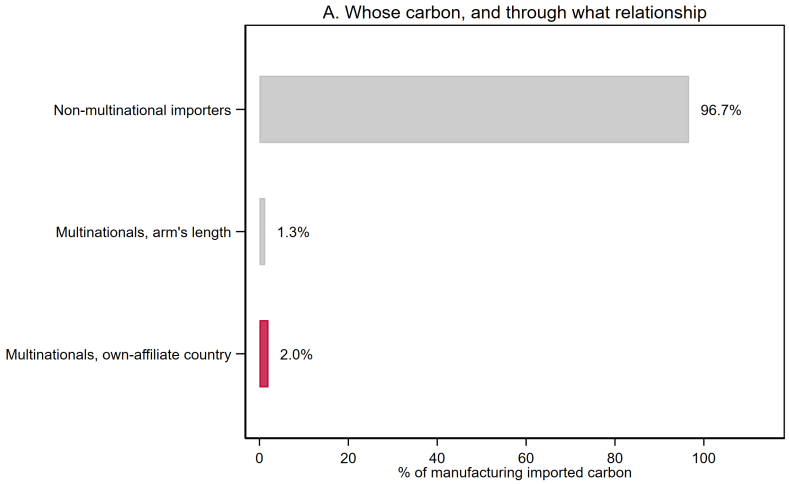
Notes. Firm and year effects. The Kleibergen–Paap rk Wald F is reported beside each row; where it falls below 10 the conventional interval is not valid and the Anderson–Rubin set, whose coverage does not depend on the first stage, is reported instead. Own-affiliate and arm's-length sourcing are the *same* firms and the *same* instrument with a different regressor: two just-identified regressions, **not** a decomposition. The own-affiliate row has $F = 0.16$ and an AR set covering the whole search grid, and is reported to show that this margin is uninformative, not null. The Anderson–Rubin sets are obtained by inverting the test on 81 points spanning $[-10, 10]$, so a set written $\supseteq [-10, 10]$ is one the test rejects nowhere on that grid; no value in it is excluded and no value outside it was examined. Panel B crosses the fixed effects with multinational status, so the difference is tested. It is 0.279 with a standard error of 0.493.

That is a statement about which margin moves. Three further pieces of evidence say the ownership margin is small, and they are worth separating because they demand different things of the data.

The first demands nothing at all. Figure 14 divides all manufacturing imported carbon by the

relationship it arrives through. Only 2.0 percent enters through a country where the importing firm owns a plant. This is a share of a flow, not an estimate: it has no first stage and no identifying assumption, and it caps the intra-firm channel at one-fiftieth of the carbon before any regression is run. It is also an upper bound, because the register records the origin country and not the counterparty, so an arm's-length purchase from an unrelated supplier in a country where the firm happens to own a plant is counted inside the 2.0 percent. The true intra-firm share is smaller.

Figure 14: Whose carbon, and through what relationship



Notes. All manufacturing imported carbon, 2008–2014. Only 2.0 percent enters through a country where the importing firm owns a plant, and that is an *upper bound*: the register records the origin country, not the counterparty, so arm's-length purchases from third parties in that country are counted here.

Table 4: Does owning a plant abroad change what a firm sources from there?

Specification	N	Coefficient
Carbon sourced (firm-year FE only)	159,839	1.076***
		(0.204)
Carbon sourced (+ origin-year FE)	159,610	0.561***
		(0.184)
Import value (+ origin-year FE)	159,610	0.580***
		(0.181)
Carbon per krona (+ origin-year FE)	159,610	-0.019
		(0.023)

Notes. Firm $\times$ year $\times$ origin. Within a firm-year, across the origins it buys from, the regressor is an indicator for the firm owning a plant in that origin. Origin-year effects absorb anything common to a country in a year. **Ownership raises how much a firm sources from an origin, not how dirty it is.** Carbon sourced rises 0.561 and import value 0.580; since $\ln(C/M) = \ln C - \ln M$, carbon per krona is their difference, -0.019 with a standard error of 0.023. That interval rules out even a six percent change in sourcing dirtiness.

The second asks whether ownership makes sourcing dirtier, and it does not need the instru-

ment to answer. In Table 4 I compare, within a firm-year and across the origins that firm buys from, the origins where it owns a plant against those where it does not, with origin-year effects absorbing anything common to a country in a year. Owning a plant in an origin raises the carbon a firm sources from there by 0.561 and raises the value it sources by 0.580. These are the same number. Since $\ln(C/M) = \ln C - \ln M$, carbon per krona is their difference, -0.019, estimated tightly enough to exclude even a six percent change in the dirtiness of what the firm buys. Ownership is endogenous and I do not treat this as the effect of a randomly assigned plant; the claim is narrower and does not require that. Whatever leads a firm to own a plant in a country, it does not lead the firm to buy dirtier things from that country.

The third is a design that does identify, and its limit is power, not bias. Figure H.1 estimates what happens to a Swedish parent's emission intensity around its first affiliate in a new country. Staggered entry with an absorbing treatment is where two-way fixed effects misbehave, but this setting has a genuine never-treated group, the firms that never open an affiliate, so the Callaway–Sant'Anna estimator applies with clean controls, and I report it alongside Sun–Abraham and two-way fixed effects. Pre-trends are flat, with $F = 1.06$ and $p = 0.348$, and no horizon separates from zero. Appendix H extends this to every horizon the panel supports and explains why the long ones are not comparable with the short ones.

What that null does and does not establish turns on the horizon. Relocating production is slow, and roughly twelve hundred treated firm-years inside a window ending in 2014 can rule out a parent that cleans up quickly after opening a plant abroad; it cannot rule out one that cleans up over fifteen years. The scope is worth stating because it is also the point: over the decade this paper observes, the margin that moves is trade, and ownership would have to work on a timescale the panel cannot see to account for what happened.

One cut of the evidence genuinely fails, and it fails for the reason Section F gave. Splitting the main estimate by multinational status leaves multinational parents at -0.249 against -0.528 for everyone else, but the multinational first stage is an F of 6.86 and the difference is tested at 6.59, so its Anderson–Rubin set runs from -1.85 to 0.89. There are 322 multinational firm-years. Cutting the shift-share by firm type destroys the identifying variation exactly as cutting it by origin group did, and the comparison settles nothing in either direction. The instrument recovers an average and will not be subdivided; the evidence on ownership has to come from designs that do not ask it to, which is what the three pieces above are.

Taken together the picture is not a collection of nulls. Arm's-length sourcing responds on an interval that excludes zero. At most two percent of the carbon arrives through an owned origin. Ownership moves how much a firm buys and not how dirty it is. And opening a plant abroad

does not clean the parent within the decade observed. A leakage policy keyed to who owns the foreign plant would address a small corner of the flow on a slow timescale, while a policy that prices the carbon embodied in imports addresses the margin that moved.

8 Conclusion

This paper examines how offshoring changes the emissions recorded at home when imports are measured by their embodied carbon instead of their monetary value. Using Swedish manufacturing firm data and a shift-share instrument based on foreign export-supply shocks, I estimate that a 10 percent increase in embodied-carbon imports reduces domestic CO_2 emissions per unit of value added by 6.7 percent. The result establishes a decline in production-based emissions intensity, but not necessarily in absolute emissions or the global emissions associated with production. Consistent with this distinction, the estimated effect on emissions in levels is statistically indistinguishable from zero.

The decline is not a technique effect. It is less energy per unit of output, not cleaner fuel: no fuel share moves and emissions per unit of energy do not change, because these firms are already almost entirely electric. Value added rises at the same time, so part of the fall in the ratio is its denominator growing, not its numerator shrinking.

The results qualify the conventional pollution-haven account of offshoring. Of the margins through which a shock to embodied carbon can travel, volume carries four fifths; the response of sourcing-origin carbon intensity is on average insignificant, as earlier firm-level work reports and as I reproduce. That average masks heterogeneity: the origin-dirtiness margin responds among firms with a broader baseline product scope, which have more room to reallocate sourcing across origins, and not among firms with narrower scope. That is what the fixed cost of reaching a new origin implies: only firms already sourcing widely have somewhere to reallocate to. What the evidence supports is a conditional origin-substitution mechanism, not a uniform shift toward dirtier production locations.

Finally, the emissions associated with foreign sourcing extend beyond domestic production. Imported goods embody 1.64 times the carbon that producing the same basket in Sweden would have released, while firms' own transport emissions and international freight rise with imports. These components are not fully captured by Swedish manufacturing's production-based footprint, and the freight is not even a relocation: carbon embodied in an imported input would have been emitted by someone either way, while the fuel burned carrying it across an ocean is emitted only because the production moved. The ownership evidence also suggests that a policy focus

limited to foreign affiliates would cover only a small share of imported manufacturing carbon: at most 2 percent enters through origins where the importing firm owns a plant. This is an upper bound based on origin and ownership data, not a direct measure of intra-firm trade.

So the ledger and the world disagree. Every production-based inventory records an improvement in Swedish manufacturing, and the carbon those inventories do not count rises. For policy the implication is narrow but firm: an instrument that prices the carbon embodied in imports addresses the margin where the displacement occurs, whereas rules that track corporate ownership reach a small corner of the flow. How large a gain such an instrument would deliver this design does not answer, since that requires a general-equilibrium model of trade and emissions and the accounting here is partial equilibrium throughout (Farrokhi and Lashkaripour, 2021; Nordhaus, 2015). What this paper supplies is the firm-level fact those exercises need: the carbon that leaves a rich country's manufacturing sector leaves on a ship, not inside a subsidiary.

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Appendix

A Variable definitions and summary statistics

Table A.1 defines every variable that enters the estimates, with its source register and, where the variable is constructed, its formula.

Table A.1: Variable definitions

Variable	Definition and source
Outcome	
CO_2, tonnes	Firm's own stationary combustion, from fuel quantities by type valued at fuel-specific emission factors. SCB energy register.
Value added, SEK	Output less intermediate inputs. SCB firm register.
Emission intensity	$\ln(CO_2/\text{value added})$. The main results also report sales and employment as alternative denominators.
Endogenous regressor and instrument	
Embodied carbon imports, C^{for}	$\sum_{c,k}$ trade value$_{ickt} \times e_{ckt}$, where e is the EXIOBASE total (direct plus upstream) CO_2 intensity of product k in country c, obtained from the Leontief inverse $S(I-A)^{-1}$.
Shift-share instrument	$\sum_k s_{ik,2005}\,\Delta\text{WES}_{kt}$: pre-sample product shares times the change in world export supply of k, leaving out Sweden.
Decomposition margins	
Import value, M	Customs value of the firm's imports. Volume margin.
Swedish-valued basket, C^{swe}	The same import basket valued at Swedish emission intensities. Product mix is $\ln(C^{swe}/M)$.
Origin dirtiness	$\ln(C^{for}/C^{swe})$: how much dirtier the chosen origins are than Sweden would be for the same basket.
Domestic technique	Emissions per unit of output on the firm's own production. Placebo margin.
Freight, E^{FR}	$\sum_c M_{ict}\, d_c\, \varphi_{m(c)}$: import weight in tonnes times CEPII distance times a modal CO_2 factor. Appendix O.
Moderators and controls	
Sector carbon intensity	Emissions per unit of value added at the 3-digit sector level, split at the manufacturing median.
EU ETS	Firm holds emission allowances in the Union registry in a given year.
Carbon tax rate	Firm's effective rate: statutory rates by fuel weighted by the firm's own fuel mix. Endogenous to the sourcing decisions under study and reported only as a specification check.
Multinational parent	Firm reports at least one foreign affiliate in the FDI register.
Own-affiliate imports	Imports from an origin in which the firm reports an affiliate. The register records origin country, not counterparty, so this is an upper bound on intra-firm trade.
Productivity	Log TFP, estimated following Wooldridge (2009).

Notes. Swedish manufacturing firms importing intermediate inputs, 2006–2014, with 2005 held out to form the instrument's baseline shares. All monetary variables are deflated to 2010 SEK. The EXIOBASE intensity e is published per million euro of final demand, so C^{for} is a mixed unit; Appendix O gives the conversion to kilograms used wherever a physical quantity is required.

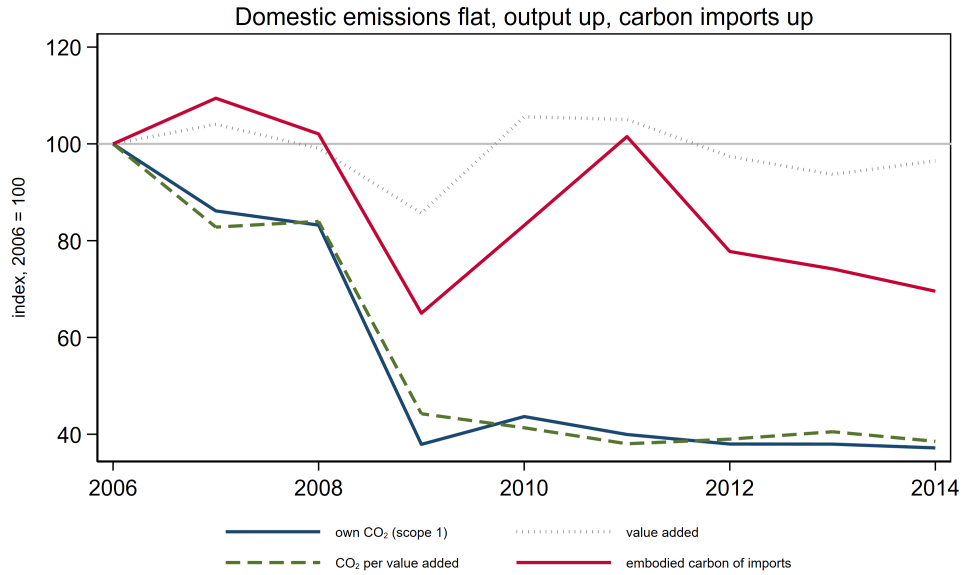
Table A.2: Summary statistics, estimating sample

Variable	N	Mean	SD	p10	Median	p90
Firm characteristics						
Employees	15,749	168.82	767.05	13.00	47.00	303.00
Sales, MSEK	15,749	643.71	4332.29	21.30	95.66	879.00
Value added, MSEK	15,749	171.14	1145.71	7.60	28.81	234.66
Energy supply, GJ	15,749	280845.90	2346319.20	1151.28	7562.60	138920.22
Distinct EXIOBASE products, first	15,749	5.06	4.21	1.00	4.00	11.00
Distinct origin countries, first y	15,749	6.99	7.18	1.00	5.00	16.00
Emissions						
CO2 scope 1, tonnes	15,749	5224.60	93386.99	6.90	76.48	1960.04
CO2 scope 2, tonnes	15,749	1152.79	6889.71	8.45	71.81	1022.79
CO2 per SEK value added, kg	15,749	0.02	0.11	0.00	0.00	0.03
CO2 per SEK sales, kg	15,749	0.00	0.04	0.00	0.00	0.01
CO2 per employee, kg	15,696	12522.98	85806.47	160.57	1851.61	16581.51
Trade and carbon imports						
Intermediate imports, MEUR	15,749	10.06	77.80	0.00	0.62	13.53
Embodied carbon of imports, tonnes	15,749	7066.54	45992.71	1.41	400.66	10471.59
Import intensity, kg CO2 per EUR	15,749	0.81	0.85	0.30	0.61	1.46
Swedish counterfactual, kg CO2 per	15,749	0.43	0.30	0.20	0.37	0.77
Foreign / Swedish intensity	15,749	2.18	3.02	0.87	1.64	3.67
Freight CO2, tonnes	15,749	400.09	2460.29	0.01	9.25	572.54
Freight as % of imported carbon	15,749	4.97	7.83	0.21	2.39	12.04
Basket churn vs own first year	15,749	0.26	0.37	0.00	0.04	1.00

Notes. Estimating sample, 2005–2014. Levels, not logs, so that magnitudes can be checked against external benchmarks: EXIOBASE embodied intensities for manufacturing run roughly 0.3–1.5 kg CO_2 per euro. Product and origin counts are taken in the firm's first sample year, before any adjustment to the shocks used for identification. Basket churn is the share of the firm's import value in products it did not buy in its own first sample year.

Figure A.1 plots emissions and output over the estimating sample year by year, which places the firm-level results against the aggregate they sit inside. Emissions fall while output rises across the window, which is the aggregate pattern the paper sets out to account for. Table A.3 then compares the firms that enter the estimating sample with those that do not, since the sample is restricted to importers of intermediate inputs and that restriction is not random: entering firms are larger and more carbon-intensive, which bounds the population the estimates describe.

Figure A.1: Emissions and output in the estimating sample



Notes. Totals over the estimating sample, by year.

Table A.3: Estimating sample against manufacturing firms outside it

Variable	In sample	Not in sample	Norm. diff.
Employees	168.82	45.29	0.213
Sales, MSEK	643.71	119.62	0.162
Value added, MSEK	171.14	30.45	0.164
CO2 scope 1, tonnes	5224.60	1173.08	0.055
CO2 per SEK value added, kg	0.02	0.01	0.021
Energy supply, GJ	280845.90	61839.86	0.119
Ever in the EU ETS	0.06	0.01	0.257

Notes. Normalised differences (Imbens and Rubin, 2015), not t-tests: with roughly 40,000 firm-years every difference is significant at 1%, so significance stars carry no information about whether the two groups resemble one another. A normalised difference above 0.25 in absolute value conventionally indicates substantial imbalance. The estimating sample is larger and more energy-intensive by construction, since it consists of importing manufacturers. Emission *intensity* — the outcome — is close to balanced. EU ETS participation is the one variable that crosses the threshold; Section 6.3 reports the main specification excluding ETS firms.

B Environmental policy stringency

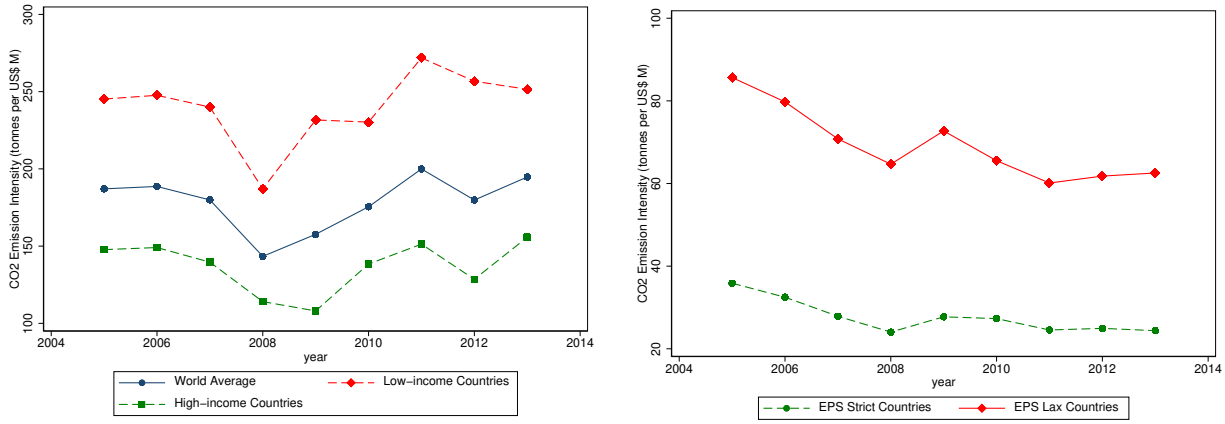
Origins are grouped by the OECD Environmental Policy Stringency index, a composite of roughly fifteen market and non-market instruments covering forty economies. I average it over 2005–2014 and measure each origin as a distance from Sweden's own value. About 80 percent of countries fall below Sweden, so a binary lax indicator separates almost nothing; I therefore cut the distance into four groups. Within half a mean deviation of Sweden is strict, half to one and a half is quite lax, and beyond that is very or extremely lax, giving a rank in $\{0, 1, 2, 3\}$.

Figure B.2 sets each origin's whole-economy carbon intensity against that grouping. Mean intensity rises across all four groups, from 109 tonnes of CO_2 per million euro of output among the strict countries to 463 among the extremely lax. Across individual countries the spread is wider still: Switzerland is the cleanest and Russia the dirtiest, a ratio of 14.5 to one, and Copeland et al. (2022) report a similar order of magnitude on different data.

The grouping is a crude summary and the figure shows where it fails. Brazil and Turkey sit in the extremely lax group and are comparatively clean, while Greece is only quite lax and is the sixth dirtiest country in the sample. Energy mix, industrial composition and income all move with policy stringency, and none of them is separated here. That is why the classification describes the origins and identifies nothing: splitting the instrument by group leaves no first stage, as Table F.1 reports, and the policy question is answered instead through the origin-dirtiness margin in Section 7.4. What this appendix establishes is weaker and sufficient: origins differ enough in carbon intensity that where a firm buys can matter for the carbon embodied in what

it buys.

Figure B.1: Trend of emission intensity by income groups and level of environmental laxity

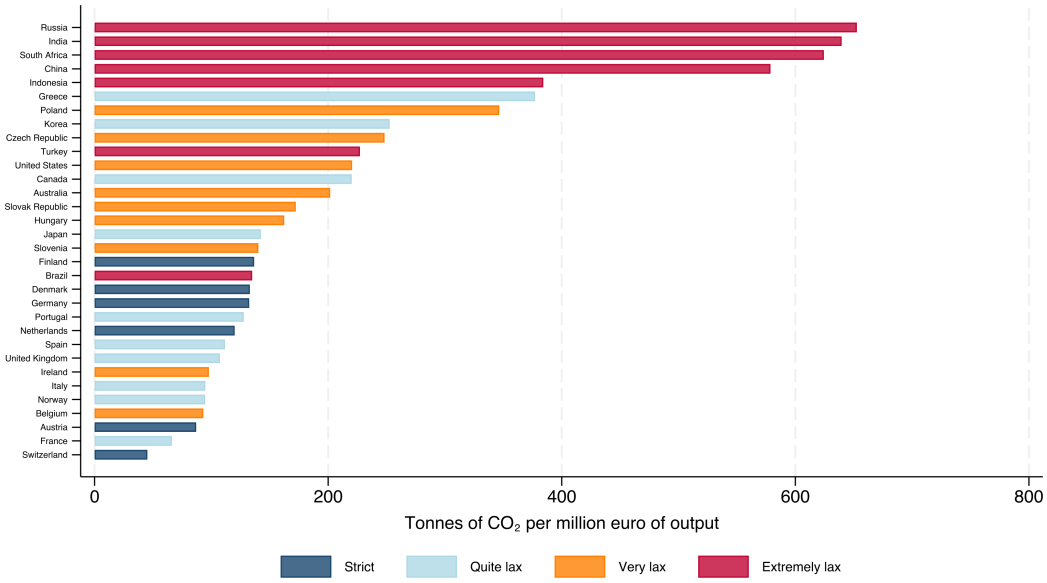


A. Low vs High Income Countries

B. EPS Lax vs EPS Strict Countries

Note: This figure plots the yearly CO_2 emission intensity (in tons per millions of US$). Low income countries includes lower middle income countries identified by the World Bank. EPS Lax and Strict countries are identified using the EPS index (see Appendix B).

Figure B.2: Emission intensity and environmental policy stringency



Notes. Each bar is a country's whole-economy CO_2 emissions divided by its output, in tonnes per million euro, averaged over 2006–2014 and summed across all products. Colours give the four-way policy grouping described in the text, built from the OECD Environmental Policy Stringency index averaged over 2005–2014 and measured as a distance from Sweden. The grouping is a country characteristic rather than a firm-level treatment, and nothing in the paper's identification rests on it. *Sources:* EXIOBASE 3 and the OECD.

C The aggregate decomposition

Decomposition of Swedish Emission. To understand the role of trade on emissions, I decompose the manufacturing CO_2 emission into three potential channels: (i) changes in the size of the manufacturing sector – scale effect, (ii) changes in emissions due to technology used in production within industries – technique effect, and (iii) changes in the mix of sectors towards less pollution-intensive industries – composition effect. [4] More formally, the CO_2 emissions from the manufacturing sector in year t, E_t can be written as follows:

$$E_t = \sum_j E_{i,t} = \sum_j v_{i,t} z_{j,t} = V_t \sum_i \theta_{j,t} z_{j,t} \tag{9}$$

where $j = 1, \ldots, n$ indexes manufacturing industries (defined at SIC 4-digits). V_t is the manufacturing monetary value of output, $\theta_{j,t}$ is each industry's share in total output $\left(\theta_i = v_{j,t}/V_t\right)$, and $z_{j,t}$ is each industry's emissions coefficient measured as the amount of pollution per monetary value of output in industry j $\left(z_{j,t} = p_{j,t}/v_{j,t}\right)$. Thus, the total CO_2 emissions in a particular year is the scale of the sector times weighted emission per monetary value of output ($V_t \sum_i \theta_{j,t} z_{j,t}$) – where θ and z are $j \times 1$ vectors containing the output shares of each of the j sectors and their emissions intensities, respectively. Equation 9 can be rewritten in a vector notation form as $E = V\boldsymbol{\theta}'z$. By applying a total differentiation on the vector notation, I obtain:

$$dE = \underbrace{\boldsymbol{\theta}' z dV}_{scale} + \underbrace{V z' d\boldsymbol{\theta}}_{composition} + \underbrace{V\boldsymbol{\theta}' dz}_{technique} \tag{10}$$

The three terms on the right-hand side in equation 10 can be interpreted as the three primary channels or effects that explain the change in total pollution over time, namely, the scale effect, the composition effect, and the technique effect. Figure 3 reports the decomposition, indexing aggregate manufacturing CO_2 to 1 in 2006, the first year the energy data exist. The construction follows Choi et al. (2023), who combine Levinson (2009) with Melitz and Polanec (2015). Line (1) varies only the scale of the sector, holding the industry mix and every industry's emission intensity at their base-year values. Line (2) additionally lets the industry mix move, so the gap between (1) and (2) measures composition. Line (3) is realised emissions, so the gap between (2) and (3) measures technique. Choi et al. (2023) split that technique effect further, into the part contributed by surviving establishments and the part contributed by entry and exit; the Swedish

[4] My working assumption, as motivated by Copeland et al. (2022) and Choi et al. (2023) is that an increase in physical unit of output (and monetary value of output) does not scale 1-to-1 with emissions because, firm adjust their production processes which can affect emissions and value added. Subsequently, I follow the decomposition exercise of Choi et al. (2023); Copeland et al. (2022); Ustyuzhanina (2022).

energy survey does not carry that split, for the reason given in the note to Figure 3.

Figure 3 tracks aggregate manufacturing CO_2 relative to 2006. The black line isolates the scale effect, the blue line adds composition, and the green line is realised emissions, so the residual gap is technique. Scale works against the decline, since output grew over the period. That composition pushes the other way is what Ustyuzhanina (2022) also report for Swedish manufacturing over 2007–2017: the sector moved toward more pollution-intensive goods, and technique had to offset both that and the growth in output.

D Abatement investment

The abatement survey and what it covers

Statistics Sweden runs an annual survey in which firms report investment and current expenditure on environmental protection, in thousands of SEK, split by purpose: clean air, water, waste, and environmental research and development. The survey would answer a question the main text leaves open, which is whether firms that offshore carbon spend less on abating what remains at home.

It cannot, and the reason is the instrument rather than the outcome. The survey draws all manufacturing firms above 250 employees, half of those between 100 and 249, and a fifth of those between 50 and 99, reaching roughly 700 firms a year. On that subsample the shift-share has almost no power. The extensive margin—an indicator for reporting any environmental investment, defined on every surveyed firm-year and conditioning on nothing—has a Kleibergen–Paap statistic of **0.08**. Conditioning on positive investment, which is the obvious worry, is not what binds: the same outcome under the inverse hyperbolic sine of Akerman et al. (2024), which keeps the zeros and so uses the same firm-years, returns a first stage below 10 as well. Every estimate in Table D.1 Panel B carries an interval wide enough to contain a hundredfold increase and a hundredfold decrease.

I therefore report no finding on abatement in either direction. These rows are not evidence that abatement is unchanged: a first stage of 0.08 is uninformative, and the discipline that forbids treating the emissions-in-levels estimate at $F = 8.6$ as a null applies here with far more force. Akerman et al. (2024), using the same registers with a value-weighted exposure measure and a longer window, find that importing raises abatement investment. Nothing here contradicts that, and nothing here supports it.

What the main text does establish about technique rests on different and much stronger evidence. On the largest first stage in the paper, $F = 22.4$, no fuel share responds to the shock and

emissions per gigajoule do not move; the domestic-technique margin of Section 7.4 is insignificant at the preferred rung in the pooled sample and in both halves of the split. The argument that the decline is not a technique effect does not depend on the abatement survey and is not weakened by its silence.

Table D.1: What else the shock moves: productivity and abatement

Outcome	N	2SLS	s.e.	KP F
Panel A. Productivity				
Total factor productivity (ln, Wooldridge)	39,804	*0.033*†	(0.102)	9.71
Value added per employee (ln)	22,259	0.217**	(0.087)	13.12
Sales per employee (ln)	22,430	0.180***	(0.065)	13.79
Panel B. Abatement				
Reports any abatement investment (0/1)	3,217	*-2.098*†	(7.675)	0.08
Investment, levels (ln), investors only	621	*-0.108*†	(1.923)	0.68
Investment per VA (ln), investors only	594	*-0.537*†	(1.377)	1.35
Akerman asinh, F series (expenditure)	3,309	*-5.726*†	(11.378)	0.34
asinh, B series (investment)	3,217	*-13.639*†	(50.451)	0.08
Panel C. Reduced form on the shock				
Total factor productivity (ln)	39,804	0.007	(0.023)	–
Value added per employee (ln)	22,259	0.076***	(0.022)	–
Sales per employee (ln)	22,430	0.064***	(0.018)	–
Akerman asinh, F series (expenditure)	3,309	-0.753	(0.718)	–
asinh, B series (investment)	3,217	-0.818	(0.718)	–
Reports any abatement investment (0/1)	3,217	-0.126	(0.103)	–

Notes. Each row is a separate 2SLS regression of the stated outcome on log embodied carbon imports, instrumented by the shift-share, with firm, sector–year and municipality–year effects and standard errors clustered by firm. The reported F is the Kleibergen–Paap rk Wald statistic. A dagger marks a row whose first stage is below 10; its stars are suppressed. Panel A reports three measures of productivity because total factor productivity is itself estimated. Panel B runs on the SCB environmental-protection survey, discussed in the text above.

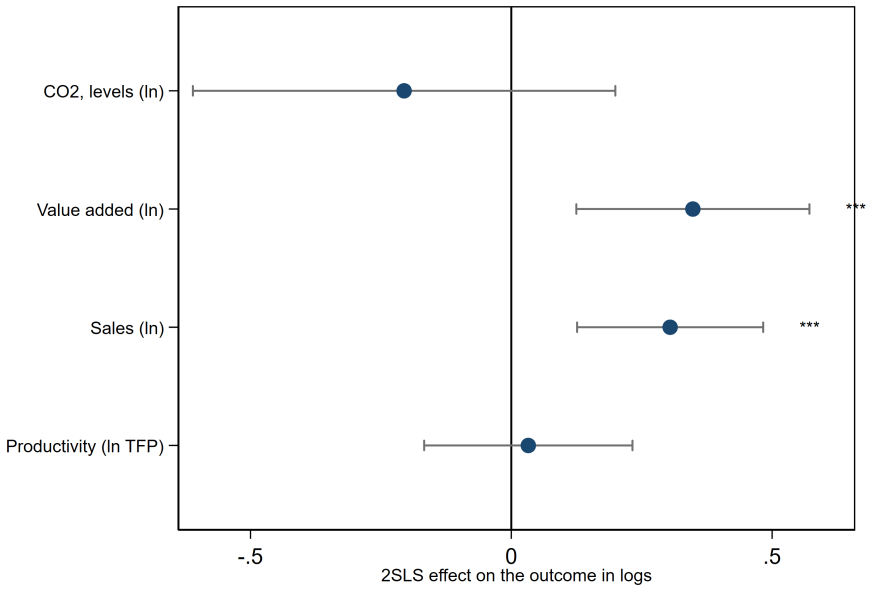
E All outcomes on the preferred specification

Section 6.5 reports the outcomes that bear on the mechanism. Figure E.1 collects all of them estimated on column (3) of Table 2, which shows the ones the text does not discuss and allows the adding-up to be checked: because $\ln(\mathrm{CO}_2/VA) = \ln \mathrm{CO}_2 - \ln VA$, the coefficient on the headline intensity must equal the coefficient on emissions in levels minus the coefficient on value added.

Two rows carry a first stage below 10 and are drawn accordingly. Emissions in levels sit at $F = 8.6$ and total factor productivity at $F = 9.71$. Neither is a null. The productivity row in particular cannot settle whether offshoring made these firms more efficient: a single estimated TFP measure on a marginal first stage is not enough, and what would settle it is the same

elasticity on several measures at once—Wooldridge TFP, value added per employee, value added per hour, and capital per worker—so that a movement in the estimated measure alone could be distinguished from a movement in the firm. That evidence is not assembled here.

Figure E.1: Numerator or denominator?



Notes. The same instrument on four other outcomes. Value added rises 0.348^{***} at $F = 13.4$. CO_2 in levels is -0.205 with a standard error of 0.207 at $F = 8.6$, and productivity is 0.033 at $F = 9.7$; both sit below the conventional threshold and neither is a null.

F The origin-group split of the instrument

The natural way to ask the question is to split the instrument by the origin's environmental policy: if arbitrage drives the result, the shock should bite through lax origins and not through strict ones.

I ran that test. It does not identify. Table F.1 reports the first stage for the pooled instrument and for the same instrument restricted to each origin group. Only the pooled version has a first stage worth the name. Restricting to EPS-lax origins gives an F of 5.6, to EPS-strict origins 0.7, and to the rest of the world 0.4, where the 2SLS coefficient of $+7.6$ is a near-zero denominator rather than an estimate. The reason is collinearity. A firm's exposure to supply shocks from the European Union is nearly the same variable as its exposure to supply shocks from everywhere, because the same baseline product shares weight both. Splitting the shock destroys the identifying variation without delivering six usable instruments. The same happens to every other cut of this instrument, including the split by firm type I come to below: it recovers an average and will

not be subdivided.

Table F.1: Splitting the instrument by origin group destroys the first stage

Shock restricted to	First-stage F	2SLS
All origins (pooled)	**18.4**	$\mathbf{-0.504}^{***}$
China / Hong Kong	7.9	-0.075
EPS-lax	5.6	-0.302
European Union	1.4	-0.287
Low income	1.1	-1.155
EPS-strict	0.7	-1.510
Rest of world	0.4	$+7.617$

Notes. Each row re-estimates column (3) of Table 2 with the shift-share built on the stated origin group alone. The reported F is the Kleibergen–Paap rk Wald statistic. **Only the pooled instrument has a first stage.** The rest-of-world coefficient of $+7.6$ divides by a first stage near zero and is not an estimate of anything. EPS classifies an origin as lax if its OECD Environmental Policy Stringency index lies below Sweden's in the same year; Appendix B describes the index.

G Heterogeneity

Two firm characteristics should matter for policy if the response is uneven: a carbon price that already binds on the firm, and a sector where carbon is a large share of costs. Neither turns out to move the estimate, and the way the test is built is what makes that statement worth anything.

Splitting a sample and comparing two confidence intervals tests nothing. I therefore cross the fixed effects with the moderator, so that the interacted model reproduces the split-sample estimates exactly while the *difference* becomes a parameter with a standard error. Table G.1 reports both moderators this way.

Neither difference is distinguishable from zero. Firms in carbon-intensive sectors give -0.575 and firms in the rest of manufacturing -0.657, a gap of 0.082. The data cannot detect whether the response depends on how carbon-intensive the firm's sector is: the difference is 0.082 with a standard error of 0.543.

The EU Emissions Trading System is the more interesting of the two and the weaker. Firms outside the system give -0.677 and firms inside it 0.433. Read naively this says the scheme switches the effect off. I do not read it that way, because the covered cell is small and its interval runs from roughly -1.35 to $+2.22$: it contains the pooled estimate, zero, and effects twice the pooled estimate in the opposite direction. This row is a statement about how few covered firms the sample holds, not a statement about carbon pricing, and I report it so that the absence of a test is on the record.

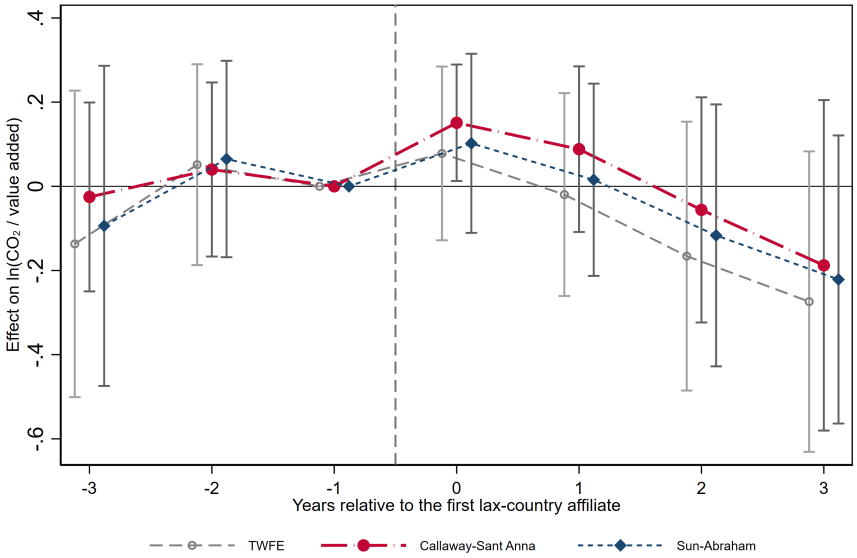
Table G.1: Heterogeneity: sector carbon intensity and the EU ETS

Moderator	Outcome	Low	High	Difference
Sector intensity	CO2 intensity	-0.657	-0.575*	0.082
		(0.421)	(0.343)	(0.543)
EU-ETS	CO2 intensity	-0.677**	0.433	1.110
		(0.294)	(0.910)	(0.956)

Notes. Fixed effects are crossed with the moderator, so the interacted model reproduces the split-sample estimates exactly and the *difference* is a tested parameter rather than a comparison of two confidence intervals. Instrument as in Table 2. **Neither difference is significant.** The ETS cell is small, so that row is a statement about power rather than about carbon pricing. Standard errors clustered by firm. $^{*}\ p < 0.10$, $^{**}\ p < 0.05$, $^{***}\ p < 0.01$.

H Opening a plant abroad

Figure H.1: Opening a plant abroad and the parent's emissions



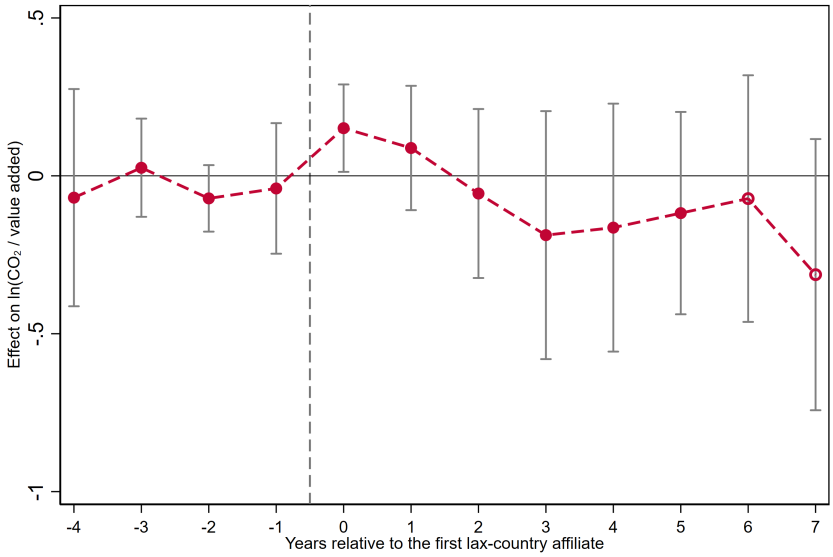
Notes. Callaway–Sant'Anna, Sun–Abraham and TWFE event studies around a firm's first affiliate in a new country, with firms that never open one as the never-treated group. Pre-trend $F = 1.06$, $p = 0.348$. Roughly twelve hundred treated firm-years, so the intervals are wide. Appendix H extends this to every horizon the panel supports and maps where Swedish affiliate employment sits.

Figure H.1 reports the response of a parent's emission intensity around its first foreign affiliate at the horizons the main text uses. Figure H.2 extends it to every horizon the panel supports. The long horizons are not comparable with the short ones: a 2005–2014 window cannot follow a late entrant far forward or an early one far back, so each horizon describes a different set of firms, and the markers resting on two cohorts or fewer are drawn hollow. The aggregates are the objects to read.

Figure H.3 shows where the affiliate employment actually sits. It is concentrated in high-

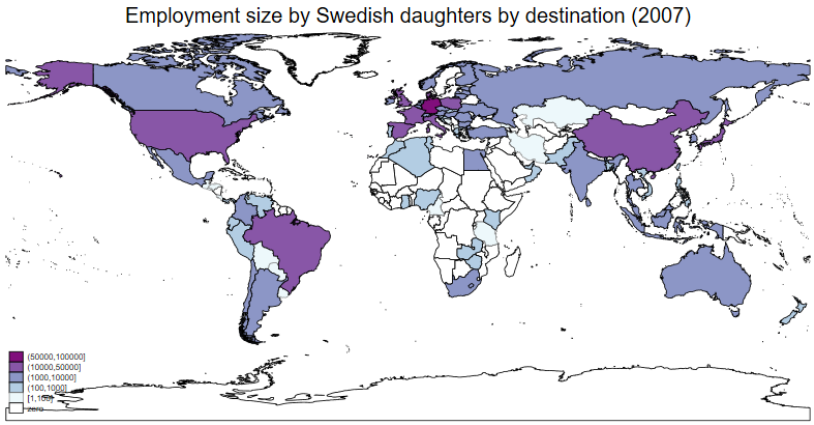
income Europe and North America, not in the destinations the pollution-haven hypothesis points to, which is a descriptive fact about the FDI register, not evidence about emissions, and is the reason the ownership channel is small before any estimation.

Figure H.2: Opening a plant abroad: longer horizons



Notes. Callaway–Sant'Anna only, at every horizon the panel supports. Hollow markers rest on two cohorts or fewer: a 2005–2014 panel cannot observe a late entrant far forward or an early one far back, so the estimate at a long horizon covers a different set of firms from the estimate at zero. Read the aggregates rather than any single horizon.

Figure H.3: Where Swedish affiliates are



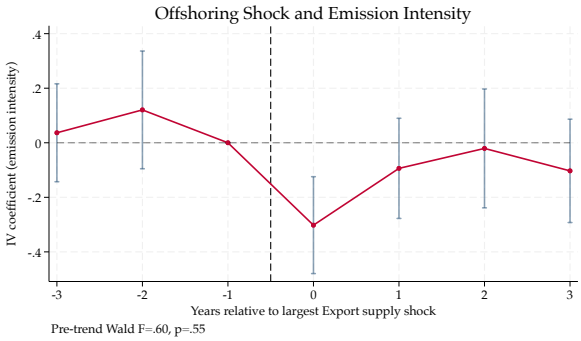
Notes. Employment in foreign affiliates of Swedish manufacturing firms. Employment rather than plants or sales because the missing and zero rate is far lower. *Source:* FDI register, not Swedish firm emissions.

I The reduced form over event time

Section 6.3 traces the instrumented response over event time. That exercise divides by the first-stage coefficient, so a sceptic can object that its shape inherits whatever the first stage is doing. The reduced form does not. Here the event-time dummies are built from the timing of the firm's largest export-supply shock, the outcome is emission intensity, and nothing is instrumented: the figure is a regression of the outcome on the instrument's timing, and its validity does not depend on the strength of the first stage.

Figure I.1 reports it, with the year before the shock normalised to zero. The leads are jointly indistinguishable from zero, with a Wald F of 0.60, so the firms that receive the largest shocks were not already on a declining path. The response begins at the shock and grows, which is the same shape the instrumented version produces. I report this in the appendix instead of the main text because it answers a narrower question than Figure 6 does—it establishes that the timing is not incidental, but it recovers no magnitude—and because it is not an elasticity.

Figure I.1: Reduced form around the largest export-supply shock



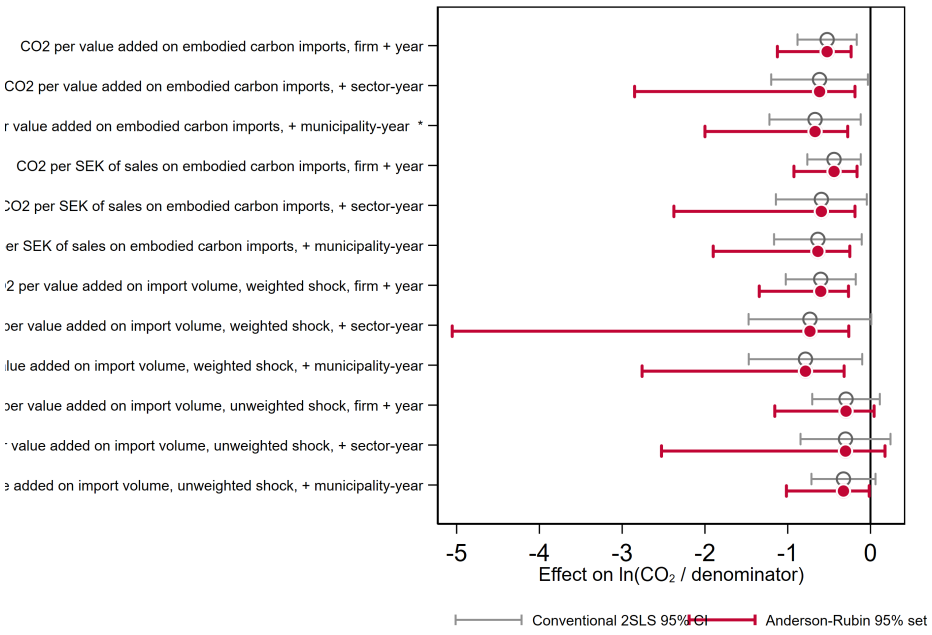
Notes. Emission intensity around the firm's largest shock, $t = -1$ normalised. Nothing is instrumented: the regressors are event-time dummies built from the *shock's* timing, so this is a reduced form. Pre-trend Wald $F = 0.60$, $p = 0.55$.

J Weak-identification-robust inference

The first stage is moderate, not strong, so the conventional two-stage-least-squares interval is not automatically trustworthy: its coverage depends on the first stage being large, and at a Kleibergen–Paap F near sixteen that is an assumption, not a fact. I therefore invert the Anderson–Rubin test over a grid of candidate values of the structural parameter and report the set of values the data cannot reject. Anderson–Rubin coverage is correct whatever the strength of the instrument, including when it is arbitrarily weak.

Figure J.1 reports those sets beside the conventional intervals, at each rung of fixed effects and for both denominators. Every carbon-weighted specification excludes zero, so the headline result does not depend on the conventional approximation, and so does the import-value specification instrumented by the unweighted shock, which is column (6) of Table 2. Column (5) needs no panel of its own: it shares an outcome, an instrument and a sample with column (3), so at $\beta_0 = 0$ the Anderson–Rubin statistic reduces to the same reduced-form regression in both, and the two columns cannot differ in whether they exclude zero. Section 6.1 draws the comparison these estimates support.

Figure J.1: Anderson–Rubin sets against conventional intervals

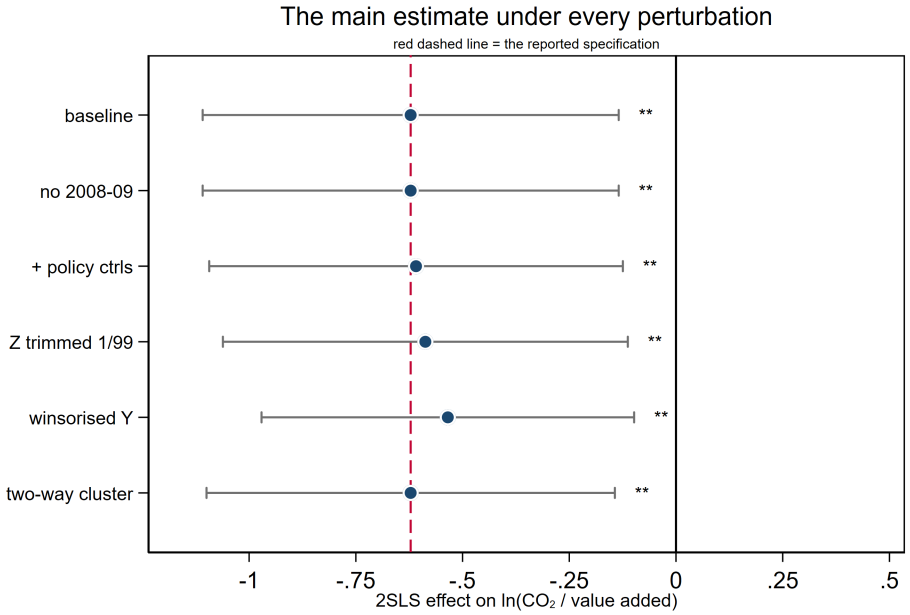


Notes. Each row is one specification, with the conventional 2SLS 95 percent interval above and the Anderson–Rubin 95 percent set below; the marker is the 2SLS point estimate in both. Sets are obtained by inverting the Anderson–Rubin test over a grid of candidate values, which is that test exactly when the model is just-identified. Rows are grouped so that the three fixed-effect rungs of a given specification sit together, and the asterisk marks column (3) of Table 2, the preferred specification. AR coverage does not depend on the strength of the first stage, so a set much wider than the interval above it means that interval was borrowing strength the instrument does not have.

K Specification choices in the shift-share design

The preferred column rests on three kinds of choice: which fixed effects absorb the variation, how the standard errors are clustered, and how emissions are scaled. Each is defensible. None is implied by the design, and until now none had been varied on its own.

Figure K.1: Robustness of the IV estimate

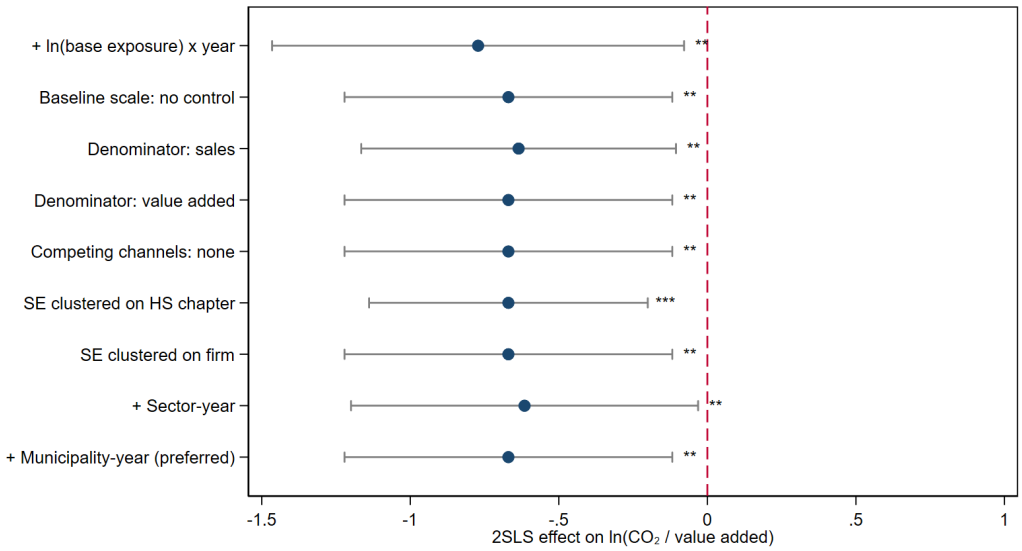


Notes. Baseline; dropping 2008–09; adding policy controls; the instrument trimmed at the first and ninety-ninth percentiles; the outcome winsorised; and two-way clustering on firm and sector–year. $^{*}\ p < 0.10$, $^{**}\ p < 0.05$, $^{***}\ p < 0.01$.

In Figure K.2, I re-estimate column (3) of Table 2 under nine such choices, one at a time. Every interval excludes zero. Absorbing sector–year effects without municipality–year effects gives -0.615. Interacting baseline exposure with year moves the estimate further from zero, not towards it, so the result is not an artefact of firms that started the period heavily exposed. Clustering on the originating HS chapter instead of the firm tightens the interval, which is what a shock common within a product chapter and not within a firm should do. The elasticity survives each choice separately.

The rung with firm effects alone is the exception, and it is omitted from the figure. Its interval runs from roughly -25 to $+8$. Without sector–year effects the instrument has too little usable variation to say anything, so that rung is uninformative about the elasticity, not evidence against it, and plotting it would compress every other row into a sliver.

Figure K.2: The carbon-import elasticity across specification choices

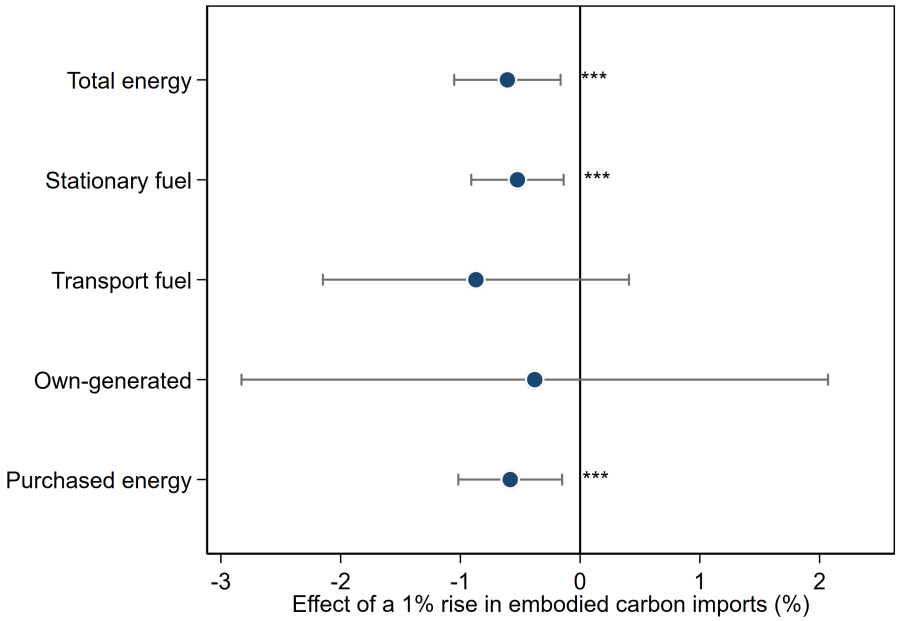


Notes. Each row re-estimates column (3) of Table 2 changing one specification choice and holding the rest fixed. The marker is the two-stage-least-squares point estimate and the horizontal line is its 95 percent interval; stars beside the interval denote significance at the 10, 5 and 1 percent levels. The dashed line marks a zero elasticity. Rows that impose a choice the preferred specification already makes reproduce it exactly, and are kept so that the full set of choices examined is visible rather than only those that move the estimate. The rung with firm effects alone is omitted from the figure: its 95 percent interval runs from about -25 to $+8$, which is uninformative about the elasticity rather than evidence against it, and on a shared axis it compresses every other row.

L The energy identity on quantities

Figure 8 splits the headline elasticity into energy per unit of output and the carbon content of that energy, both as intensities. Figure L.1 runs the same split on the quantities underneath: total energy in gigajoules and total CO_2 in tonnes. Energy is what the SCB survey measures directly and CO_2 is constructed from it by fuel-specific emission factors, so the two rows add exactly and the figure is a check on the construction, not independent evidence.

Figure L.1: Energy and carbon components



Notes. Energy is the primitive the survey measures and CO_2 is constructed from it by fuel-specific factors, so this split is an identity rather than a second estimate.

M Robustness of the decomposition

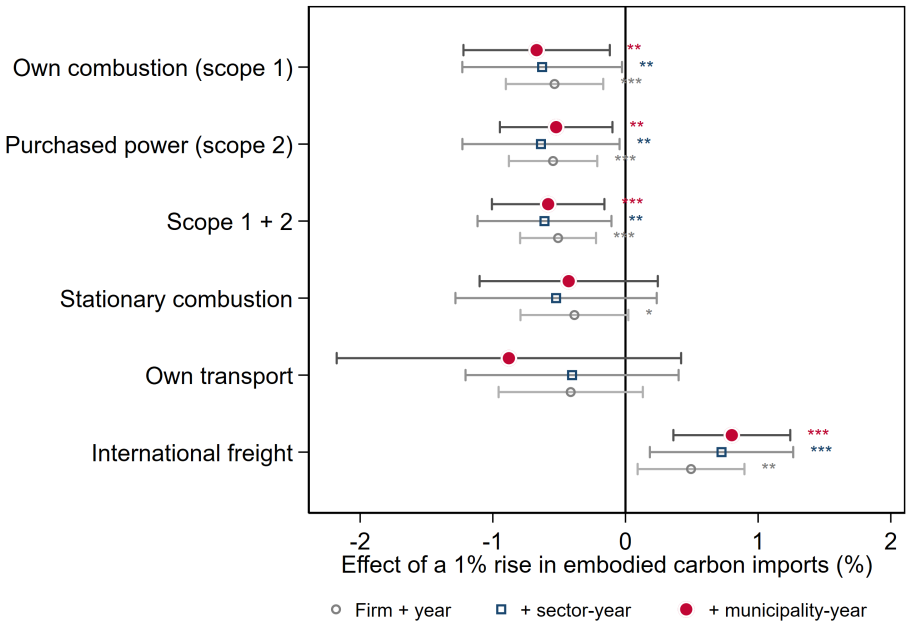
Figure M.1 repeats the component decomposition at each of the three fixed-effect rungs. The purpose is not to find the specification that gives the largest coefficient but to show which conclusions survive all three and which do not, since a result that appears only at the loosest rung is a result about what the looser specification fails to absorb.

N The aggregate decomposition

Swedish manufacturing cut its emission intensity by 94 log points between 2006 and 2014. In Table N.1 I split that change into the part occurring inside continuing firms and the part due to reallocation between firms together with entry and exit, and in Figure N.1 I set the within-firm term against the change the estimated elasticity attributes to offshoring.

The two panels of the table measure the same decline on different scales and do not agree about it. In log points, continuing firms account for 91.5 percent of the fall. In levels they account for 48 percent, and net entry for 68 per cent, with reallocation between survivors working the other way. A level is dominated by the largest and dirtiest plants and moves sharply when a few of them close, which is what drives the two apart. I report the log decomposition in the body

Figure M.1: Components across fixed-effect rungs



Notes. The decomposition of Figure 7 re-estimated at successively richer fixed effects.

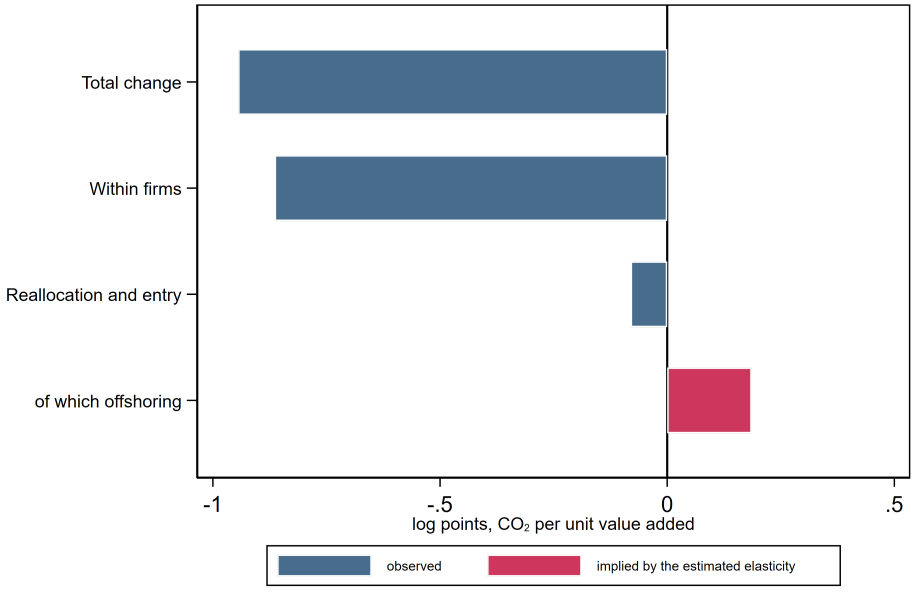
because the quantity attributed to offshoring is itself a log-point quantity: it is formed as the value-added-weighted sum across firms of $\hat{\beta}$ times each firm's change in log embodied carbon imports. It is commensurable with Panel B and not with Panel A.

The attributed quantity carries the opposite sign because of the regressor rather than the elasticity. Among firms observed in both the first and the last year of the window, the value of what they imported rose by 21 log points while the carbon embodied in it fell by 11, so carbon per krona of imports fell by 32. Firms bought more from abroad, and what they bought grew cleaner faster than the volumes grew.

Nothing in the construction forced that answer. Had the carbon embodied in these firms' imports grown with the value of those imports, the same elasticity over the same window would have attributed a substantial part of the within-firm decline to offshoring. The attribution runs against the decline because carbon per krona of imports fell, not because the design cannot detect the channel: column (3) of Table 2 estimates that channel on an interval excluding zero. The magnitude here is a fact about what Swedish firms bought over this decade, not a property of the method.

That attributed quantity asks the estimated elasticity to do something the research design does not identify. I estimate $\hat{\beta}$ within sector–year and municipality–year cells, so it compares firms facing common aggregate conditions and sets aside precisely the movement over time

Figure N.1: The decline in emission intensity, and what offshoring accounts for



Notes. Log points of CO_2 per unit of value added, Swedish manufacturing, 2006–2014. The first three bars decompose the observed change into the part occurring inside continuing firms and the part due to reallocation between firms together with entry and exit; they sum to the total by construction. The fourth bar is not observed. It is the change the estimated elasticity attributes to offshoring, formed as the value-added-weighted sum across firms of $\hat{\beta}$ times each firm's change in log embodied carbon imports, with $\hat{\beta}$ taken from column (3) of Table 2. It carries the opposite sign to the decline because embodied carbon imports fell over this window. The text states what carrying the elasticity to the aggregate requires.

that the decomposition is about. Carrying it to the aggregate requires that the relative effect survives aggregation: that one firm's offshoring does not displace another firm's emissions, and that input and output prices do not adjust. The exercise is therefore accounting under a stated assumption rather than a counterfactual. The conclusion in the body survives the qualification, because the attributed effect does not merely fall short of the decline. It carries the opposite sign, and the assumption would have to fail both severely and in one particular direction to reverse that. The required failure is specific. Displacement would have to exceed the attributed effect in magnitude and reverse its direction: about one and a half times that effect to account for a tenth of the within-firm decline, and between five and six times to account for all of it.[5]

[5]From Table N.1, with log changes scaled by one hundred: the within-firm decline is 86.4 log points and offshoring is attributed 18.6 in the opposite direction. A tenth of the decline is 8.6, so displacement must deliver $8.6 + 18.6 = 27.2$ log points, which is 1.5 times the attributed effect; the whole decline requires 105.0, which is 5.6 times.

Table N.1: The decline in emission intensity, 2006–2014

Panel A. Levels: kg CO_2 per SEK of value added		
	Change	Share of total
Total, 2006–2014	-0.032473	100.0
within firms	-0.015464	47.6
between firms	0.005023	-15.5
net entry	-0.022032	67.8
Panel B. Log points		
	Change	Share of total
Total, 2006–2014	-0.9444	100.0
within firms	-0.8640	91.5
reallocation and entry	-0.0804	8.5
Implied by the estimated elasticity, same firms		
within term, these firms	-0.8640	
attributed to offshoring	0.1859	

The implied effect and the within-firm decline have **opposite signs**: embodied carbon imports fell over this window, so a negative elasticity implies offshoring raised intensity. A share of the decline is therefore not defined and none is reported.

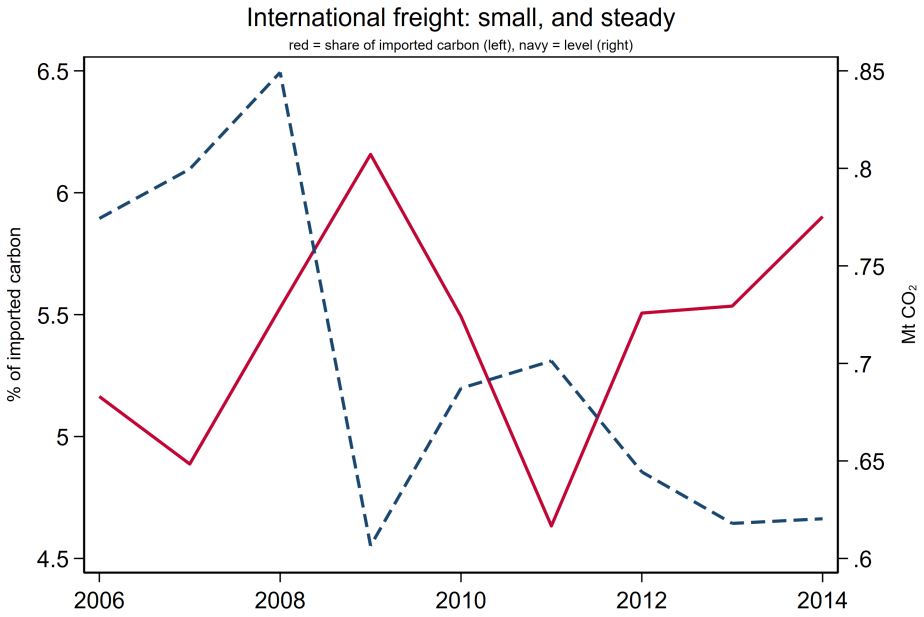
Notes. Panel A measures the change in kilograms of CO_2 per krona of value added; Panel B measures it in log changes, which the text quotes as log points after multiplying by one hundred, so the -0.9444 in the first row of Panel B is the 94-log-point fall discussed above. Within-firm, between-firm and net-entry terms sum to the total by construction in each panel. The final block applies the elasticity from column (3) of Table 2 to each firm's observed change in log embodied carbon imports and aggregates with base-year value-added weights.

O Freight-emission construction

I construct international freight emissions as $E^{\mathrm{FR}}_{it} = \sum_c M_{ict}\, d_c\, \varphi_{m(c)}$, where M_{ict} is the weight (in tonnes) of firm i's imports from country c, d_c is the CEPII population-weighted Sweden–c distance, and φ_m is the CO_2 emission factor (grams per tonne-kilometre) of transport mode m. Central modal factors follow Cristea et al. (2013) and the GLEC framework: sea 15, rail 22, road 100, and air 600 g CO_2 per tonne-km. Mode is assigned by geography—road for contiguous or short-haul intra-EU routes, sea otherwise, with air reserved for high value-to-weight flows. Within this construction, distance varies across origins by more than an order of magnitude and the modal factors span a factor of forty, so mode assignment matters at least as much as distance for E^{FR}. Whether the estimated freight response survives a sweep over the plausible ranges (sea 8–40, road 60–160) and over alternative mode assignments is not established here, and I therefore make no robustness claim for it.

Figure O.1 reports the resulting series, in levels and as a share of the firms' own reported emissions.

Figure O.1: International freight emissions: level and share

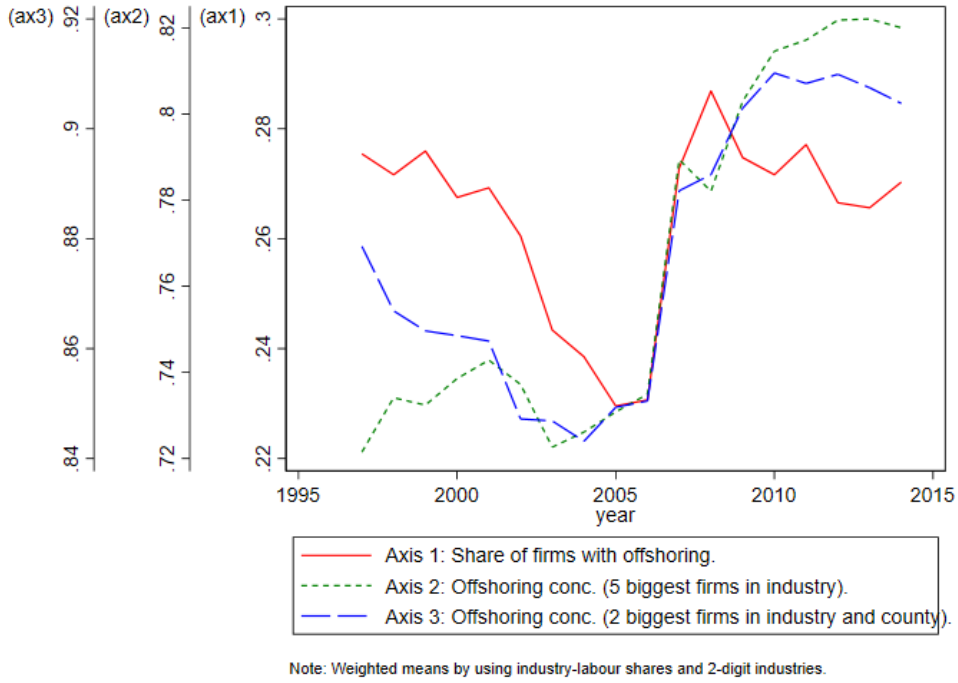


Notes. The dashed line is total international freight CO_2 carrying Swedish manufacturing imports, in megatonnes, against the right axis. The solid line is the same quantity as a percentage of the carbon embodied in those imports, against the left axis. The level falls across the window while the share rises, because embodied carbon imports fell faster than the freight carrying them. Construction is as given above; mode is inferred from geography rather than observed and import weights are net mass, so both series are lower bounds.

P Concentration of carbon offshoring

Section 4 describes how concentrated carbon offshoring is across firms within an industry. Figure P.1 reports the three series that description rests on.

Figure P.1: Concentration of carbon offshoring within 2-digit industries, 1997–2015



Notes. This figure provides insights into the pollution offshoring patterns at the sector level by illustrating the concentration of offshoring within 2-digit industries. The concentration is measured in terms of the offshoring share represented by: (i) the offshoring activities of all firms engaged in offshoring within each industry; (ii) the total offshoring conducted by the five largest offshoring firms within each industry; and (iii) the total offshoring conducted by the two largest firms within each industry and county. Weighted means, using industry–labour shares and 2-digit industries. The three series are plotted against three separate vertical scales, shown at the left, so their levels are not directly comparable across series.